A Certified and Differentiable Parametric Solver for Binary-Black-Hole Puncture Initial Data

Frederik De Ceuster[1, 2, 3, *] and Tjonnie G.-F. Li[1, 2, 3]

[1] *Institute for Theoretical Physics, KU Leuven, Celestijnenlaan 200D, 3001 Leuven, Belgium*
[2] *Leuven Gravity Institute, KU Leuven, Celestijnenlaan 200D, 3001 Leuven, Belgium*
[3] *STADIUS Center for Dynamical Systems, Signal Processing and Data Analytics, KU Leuven, Kasteelpark Arenberg 10, 3001 Leuven, Belgium*

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Constraint-satisfying initial data for binary black holes require the repeated solution of a nonlinear elliptic boundary-value problem throughout the physical parameter space. We present a parametric initial-data solver that combines sparse-grid spectral interpolation with Newton refinement to construct certified, differentiable initial data over a chosen family of binary configurations. Interpolation is used only to provide warm starts for the underlying elliptic solver, rather than to replace it. A small number of Newton iterations refines each interpolated solution to a constraint residual $\|R\|_\infty \leq 10^{-10}$, independent of the interpolation error, thereby certifying every generated initial-data set. A JAX implementation further provides derivatives of the certified initial data with respect to the physical parameters, enabling gradient-based parameter targeting, optimization, and sensitivity analysis. We demonstrate the method for quasi-circular binary-black-hole puncture initial data by constructing four-dimensional aligned-spin and eight-dimensional general-spin models. The resulting initial data are validated against TwoPunctures, and the method is applied to certified targeting of the physical ADM mass and angular momentum and to effective-potential eccentricity reduction using a dedicated (b, P_t) model.

I. INTRODUCTION

Every numerical-relativity simulation of a binary black hole begins by solving the Einstein constraint equations on the initial time slice. For conformally flat, maximally sliced puncture data, this is a nonlinear elliptic boundary-value problem: the Hamiltonian constraint for the conformal factor in Lichnerowicz–York form [1–4]. This problem must be solved once for each configuration.

Modern applications, however, rarely require a single solution. Parameter surveys, waveform catalogs, eccentricity reduction, and mass and spin targeting all require the same elliptic problem to be solved repeatedly across the binary-black-hole parameter space. The repeated solution of the constraint equations therefore becomes the dominant recurring computational cost [5]. The computational challenge is therefore not solving a single elliptic problem, but efficiently solving a family of closely related elliptic problems across the physical parameter space.

Reduced-order modeling is a mature response to exactly this kind of repeated computation. In gravitational-wave physics, however, it has been developed almost exclusively for *waveforms*: reduced-basis template banks and surrogate models reproduce waveforms across parameter space at a tiny fraction of the cost [6–9]. The elliptic initial-data solve that precedes every numerical-relativity evolution has largely remained outside this development. Reduced-basis emulation does exist for the neutron-star sector, for example through Tolman–Oppenheimer–Volkoff emulators over equation-of-state parameters [10], but no comparable parametric solver exists for binary-black-hole puncture initial data.

From the perspective of numerical analysis, this gap is somewhat surprising. Parametric collocation, sparse interpolation, and reduced-basis methods for parameter-dependent elliptic partial differential equations are well-established [11–15]. When the parameter-to-solution map is analytic, the interpolation error decays exponentially with the number of parameter samples at a rate determined by the nearest singularity in the complex parameter plane through the Bernstein-ellipse estimate [16]. These results suggest that binary-black-hole puncture initial data should also admit efficient parametric representations over suitable regions of parameter space.

Several recent developments have addressed complementary aspects of this problem. Mendes *et al.* [5] accelerate parameter targeting through a Broyden iteration that repeatedly re-solves the constraint equations. Tommasini *et al.* [17] use Gaussian-process regression trained on the SXS catalogue to predict low-eccentricity orbital parameters, reducing the number of trial evolutions required for eccentricity reduction. Zhou *et al.* [18] introduce a physics-informed neural-network solver for the puncture Hamiltonian constraint on a single configuration and identify parameterized initial-data solvers as an important direction for future work. Analytic perturbative constructions [19] and neural representations of spacetime metrics [20] provide additional complementary approaches. None of these methods, however, constructs a parametric solver that can certify the constraint satisfaction, and provide exact derivatives of the resulting initial data with respect to the physical parameters.

This paper addresses this gap by introducing a parametric solver for binary-black-hole puncture initial data

* frederik.deceuster@kuleuven.be

with certified constraint satisfaction and exact differentiability with respect to the physical parameters. Rather than replacing the elliptic solve with an approximation, the method constructs a sparse interpolant that provides a high-quality initial guess for the nonlinear solve, after which a small number of Newton iterations recover a fully converged solution satisfying the Einstein constraint equations. We demonstrate the approach for both a four-dimensional aligned-spin model and an eight-dimensional general-spin quasi-circular model, and apply it to two representative tasks: targeting physical parameters and reducing orbital eccentricity.

The proposed solver combines three properties that, to our knowledge, have not previously been demonstrated together for binary-black-hole puncture initial data: it is parametric, provides certified constraint satisfaction, and is exactly differentiable with respect to the physical parameters. First, it provides *certified constraint satisfaction*. Every queried parameter is refined by Newton iterations until the constraint residual satisfies $\|R\|_\infty \leq 10^{-10}$, ensuring that every returned initial datum satisfies the Einstein constraints to a prescribed tolerance. The sparse interpolant therefore accelerates the computation but does not replace the underlying elliptic solver. Second, it provides *exact differentiability*. The entire construction is implemented in JAX [21], making the resulting parameter-to-solution map differentiable with respect to the physical parameters. These derivatives are obtained automatically through algorithmic differentiation.

The remainder of this paper is organized as follows. Section II reviews the puncture initial-data formulation and the resulting nonlinear elliptic constraint equation. Section III describes the spectral elliptic solver that produces a single constraint-satisfying datum. Section IV builds the parametric solver over it, combining sparse interpolation, Newton refinement that certifies every returned datum, and a differentiable JAX implementation. Section V constructs and characterizes the four-dimensional aligned-spin and eight-dimensional general-spin quasi-circular models, including their convergence and the analyticity limits of the parameter map. Section VI applies the method to certified parameter targeting and effective-potential eccentricity reduction. Finally, Section VII discusses the limitations, implications, and conclusions of this work. Validation of the initial data against TwoPunctures is collected in Appendix A.

II. PUNCTURE INITIAL DATA

We briefly review the puncture formulation of binary-black-hole initial data used throughout this work. We adopt conformally flat, maximally sliced initial data within York's conformal transverse–traceless decomposition [1, 4], rather than the conformal thin-sandwich formulation [22]. Under these assumptions, the momentum constraint is satisfied analytically, leaving a single nonlinear elliptic equation for the conformal factor.

A. Constraint and puncture decomposition

In the conformal decomposition, the spatial metric and extrinsic curvature are $\gamma_{ij} = \psi^4 f_{ij}$ and $K^{ij} = \psi^{-10}\hat{A}^{ij}$, where f_{ij} is the flat conformal metric and $\hat{A}^{ij}$ is the conformal (Bowen–York) extrinsic curvature. We assume maximal slicing, $K \equiv f_{ij}K^{ij} = 0$. If $\hat{A}^{ij}$ is transverse with respect to the flat metric, $\partial_i \hat{A}^{ij} = 0$, the momentum constraint is satisfied analytically (Sec. II B), while the Hamiltonian constraint reduces to the nonlinear Lichnerowicz–York equation

$$\Delta\psi \;=\; -\tfrac{1}{8}\,\psi^{-7}\,\hat{A}_{ij}\hat{A}^{ij}, \tag{1}$$

where Δ is the flat Laplacian and indices are raised and lowered with f_{ij}. This is the only equation that must be solved numerically.

Equation (1) is defined on an unbounded domain and is singular at each puncture because $\hat{A}^2$ diverges there. Following Brandt and Brügmann [3], we decompose the conformal factor as $\psi = \psi_{\rm BL} + u$, where the N-puncture Brill–Lindquist background is

$$\psi_{\rm BL} \;=\; 1 + \sum_X \frac{m_X}{2r_X}, \tag{2}$$

where $r_X \equiv |\boldsymbol{x} - \boldsymbol{x}_X|$ is the coordinate distance to puncture X. The Brill–Lindquist term contains all $1/r_X$ singularities and is harmonic away from the punctures. The regular correction therefore satisfies

$$\Delta u \;=\; -\tfrac{1}{8}\,(\psi_{\rm BL} + u)^{-7}\,\hat{A}_{ij}\hat{A}^{ij}, \tag{3}$$

where, for the binary, we take $N = 2$ punctures of bare masses m_A and m_B located at $z = \pm b$. Solving Eq. (3) for u determines the complete initial data.

B. Bowen–York source term

The momentum constraint is satisfied analytically by the Bowen–York extrinsic curvature [2],

$$\hat{A}^{ij} \;=\; \sum_{X\in\{A,B\}} \big(\hat{A}^{ij}_{P,X} + \hat{A}^{ij}_{S,X}\big), \tag{4}$$

where the momentum and spin contributions are

$$\hat{A}^{ij}_{P,X} \;\equiv\; \frac{3}{2r_X^2}\Big[P^i_X n^j_X + P^j_X n^i_X - \big(f^{ij} - n^i_X n^j_X\big)(\boldsymbol{P}_X\cdot\boldsymbol{n}_X)\Big], \tag{5}$$

$$\hat{A}^{ij}_{S,X} \;\equiv\; \frac{3}{r_X^3}\big(v^i_X n^j_X + v^j_X n^i_X\big), \tag{6}$$

where $\boldsymbol{P}_X$ and $\boldsymbol{S}^X$ are the linear momentum and spin of puncture X, respectively, $\boldsymbol{v}_X \equiv \boldsymbol{S}^X \times \boldsymbol{n}_X$, and $\boldsymbol{n}_X \equiv (\boldsymbol{x} - \boldsymbol{x}_X)/r_X$ is the unit radial vector centered on puncture

X. The tensor f^{ij} in Eq. (5) is the inverse flat conformal metric; in the Cartesian coordinates used to construct the tensor, $f^{ij} = \delta^{ij}$. Each Bowen–York contribution is transverse and trace-free with respect to f_{ij}. Their sum therefore satisfies the momentum constraint analytically for arbitrary momenta, spins, and separations.

The Hamiltonian constraint (3) depends only on the scalar contraction $\hat{A}_{ij}\hat{A}^{ij}$. Defining $\hat{A}_X \equiv \hat{A}_{P,X} + \hat{A}_{S,X}$ as the total tensor for puncture X, the contraction decomposes into two self-contractions and a cross term,

$$\hat{A}_{ij}\hat{A}^{ij} \;=\; \hat{A}_A\!:\!\hat{A}_A \;+\; 2\,\hat{A}_A\!:\!\hat{A}_B \;+\; \hat{A}_B\!:\!\hat{A}_B. \tag{7}$$

Each self-contraction is

$$\begin{aligned}\hat{A}_X\!:\!\hat{A}_X \;=\; & \frac{9}{2r_X^4}\left[P_X^2 \;+\; 2(\boldsymbol{P}_X\!\cdot\!\boldsymbol{n}_X)^2\right] \\ & + \frac{18}{r_X^6}\,|\boldsymbol{S}^X\!\times\!\boldsymbol{n}_X|^2 + \frac{18}{r_X^5}\,(\boldsymbol{P}_X\!\times\!\boldsymbol{S}^X)\!\cdot\!\boldsymbol{n}_X,\end{aligned} \tag{8}$$

where the three terms are the momentum, spin, and momentum–spin contributions, respectively. The cross term is

$$\begin{aligned} 2\,\hat{A}_A\!:\!\hat{A}_B \;=\; & \frac{9}{2r_A^2 r_B^2}\,\Pi_{AB} \;+\; \frac{36}{r_A^3 r_B^3}\,\Sigma_{AB} \\ & +\; \frac{18}{r_A^2 r_B^3}\,\Xi_{AB} \;+\; \frac{18}{r_A^3 r_B^2}\,\Xi_{BA},\end{aligned} \tag{9}$$

where the momentum–momentum (Π), spin–spin (Σ), and momentum–spin (Ξ) invariants are

$$\begin{aligned}\Pi_{AB} \;\equiv\; & 2(\boldsymbol{P}_A\!\cdot\!\boldsymbol{n}_B)(\boldsymbol{P}_B\!\cdot\!\boldsymbol{n}_A) - 3(\boldsymbol{P}_A\!\cdot\!\boldsymbol{n}_A)(\boldsymbol{P}_B\!\cdot\!\boldsymbol{n}_B) \\ & + 2(\boldsymbol{n}_A\!\cdot\!\boldsymbol{n}_B)\big[(\boldsymbol{P}_A\!\cdot\!\boldsymbol{P}_B) \\ & \qquad + (\boldsymbol{P}_A\!\cdot\!\boldsymbol{n}_A)(\boldsymbol{P}_B\!\cdot\!\boldsymbol{n}_A) \\ & \qquad + (\boldsymbol{P}_B\!\cdot\!\boldsymbol{n}_B)(\boldsymbol{P}_A\!\cdot\!\boldsymbol{n}_B)\big] \\ & + (\boldsymbol{P}_A\!\cdot\!\boldsymbol{n}_A)(\boldsymbol{P}_B\!\cdot\!\boldsymbol{n}_B)(\boldsymbol{n}_A\!\cdot\!\boldsymbol{n}_B)^2, \end{aligned} \tag{10}$$

$$\Sigma_{AB} \;\equiv\; (\boldsymbol{v}_A\!\cdot\!\boldsymbol{v}_B)(\boldsymbol{n}_A\!\cdot\!\boldsymbol{n}_B) + (\boldsymbol{v}_A\!\cdot\!\boldsymbol{n}_B)(\boldsymbol{v}_B\!\cdot\!\boldsymbol{n}_A), \tag{11}$$

$$\begin{aligned}\Xi_{AB} \;\equiv\; & (\boldsymbol{P}_A\!\cdot\!\boldsymbol{v}_B)(\boldsymbol{n}_A\!\cdot\!\boldsymbol{n}_B) + (\boldsymbol{P}_A\!\cdot\!\boldsymbol{n}_B)(\boldsymbol{n}_A\!\cdot\!\boldsymbol{v}_B) \\ & + (\boldsymbol{P}_A\!\cdot\!\boldsymbol{n}_A)(\boldsymbol{n}_A\!\cdot\!\boldsymbol{n}_B)(\boldsymbol{n}_A\!\cdot\!\boldsymbol{v}_B),\end{aligned} \tag{12}$$

and where Ξ_{BA} follows from exchanging A and B.

C. Regularity of the puncture formulation

The puncture decomposition isolates the singular part of the conformal factor in the analytic Brill–Lindquist background $\psi_{\rm BL}$. Near puncture X, $\psi_{\rm BL}^{-7} \sim r_X^7$, whereas the momentum and spin self-contractions scale as $\hat{A}_{P,X}^2 \sim r_X^{-4}$ and $\hat{A}_{S,X}^2 \sim r_X^{-6}$, respectively. Hence, their contributions to the effective source in Eq. (3) vanish at least linearly as $r_X \to 0$. The cross terms are likewise regular and vanish at the puncture. Consequently, u is smooth throughout the compactified computational domain, making it well suited to the spectral discretization introduced in Sec. III.

III. CONSTRAINT SOLVER

We solve Eq. (3) for the regular correction u by spectral collocation on a single spectral patch. We use the Ansorg–Brügmann–Tichy (TwoPunctures) spectral method [23], extended with a Fourier expansion in ϕ for non-axisymmetric configurations. This solver provides the reference solution on which the parametric method of Sec. IV is built and against which it is validated.

A. The prolate-spheroidal single patch

The Ansorg–Brügmann–Tichy (ABT) coordinate system [23] maps the two-puncture domain onto a single compact spectral patch. Starting from cylindrical coordinates (ρ, z, ϕ), it introduces prolate-spheroidal coordinates whose foci coincide with the punctures at $z = \pm b$, together with a compactified radial coordinate. With $A \in [0,1]$ and $B \in [-1,1]$, the transformation to cylindrical coordinates is

$$\frac{\rho}{b} \;=\; \frac{2A\sqrt{1-B^2}}{1-A^2}, \qquad \frac{z}{b} \;=\; \frac{(1+A^2)\,B}{1-A^2}. \tag{13}$$

The coordinate edges have a clean meaning: $A = 1$ is spatial infinity, $A = 0$ is the inner axis segment $|z| \le b$, the edges $B = \pm 1$ are the outer axis $|z| \ge b$, and the two corners $(A = 0, B = \pm 1)$ are the punctures themselves.

Because the ABT coordinates are orthogonal, the flat-space Laplacian takes the separable form

$$\begin{aligned}\Delta u \;=\; & \frac{(1-A^2)^2}{b^2\,\mathcal{D}}\Bigg[\frac{(1-A^2)^2}{4A}\,\partial_A\big(A\,\partial_A u\big) \\ & \qquad + \partial_B\big[(1-B^2)\,\partial_B u\big]\Bigg] + \frac{\partial_\phi^2 u}{\rho^2},\end{aligned} \tag{14}$$

with $\mathcal{D} \equiv (1+A^2)^2 - B^2(1-A^2)^2$ and ρ from Eq. (13).

In a Fourier basis in ϕ, the azimuthal derivative diagonalizes, $\partial_\phi^2 \to -m^2$, so each Fourier mode can be treated independently. The linear Lichnerowicz operator therefore decomposes into independent two-dimensional blocks, eliminating the need to assemble the full three-dimensional dense matrix.

The regular correction is represented nodally: the unknown is the tensor of collocation values $U_{ijk} \equiv u(A_i, B_j, \phi_k)$, rather than the modal coefficients (c_{nl}^m below). This representation is equivalent to the truncated spectral expansion

$$u(A,B,\phi) \;=\; \sum_{|m|\le N_\phi/2}\;\sum_{n=0}^{N_A}\;\sum_{l=0}^{N_B-1} c_{nl}^m\;\phi_{nl}^m(A,B,\phi) \tag{15}$$

with basis functions,

$$\phi_{nl}^m(A,B,\phi) \;\equiv\; T_n(2A{-}1)\,P_l(B)(1{-}B^2)^{|m|/2}\,e^{im\phi}. \tag{16}$$

The basis consists of a Fourier expansion in the azimuthal angle ϕ, (with $c_{nl}^{-m} = (c_{nl}^{m})^*$ since u is real), Chebyshev polynomials T_n in the compactified radial coordinate A, and Legendre polynomials P_l in the angle B. The associated-Legendre prefactor $(1-B^2)^{|m|/2}$ carries the exact behaviour of the regular m-mode at the outer axis $B = \pm 1$. For odd m this yields a square-root branch point that a bare polynomial can only resolve algebraically, so we solve for the smooth polynomial factor and differentiate the prefactor analytically, restoring spectral convergence. The coefficients c_{nl}^{m} are not formed explicitly. The azimuthal FFT maps the N_ϕ node values to the modes m, and within each meridian block the A- and B-derivatives act on the nodal values through the Chebyshev and Legendre differentiation matrices.

A solved field must also be evaluated *off* the collocation grid at arbitrary coordinates (A, B, ϕ). Although this could be done by forming the coefficients c_{nl}^{m} and evaluating the expansion in Eq. (15), we instead interpolate the nodal values directly. We use the numerically stable barycentric-Lagrange formula [16] in A and B, applied to the smooth polynomial factor with the analytic $(1-B^2)^{|m|/2}$ prefactor restored for each mode, and equispaced trigonometric interpolation in ϕ. This procedure is not an approximation to the truncated spectral representation: the barycentric interpolant is the unique polynomial of the retained degree passing through the collocation values, while the trigonometric interpolant is the corresponding unique band-limited periodic interpolant. Consequently, the interpolated field is mathematically identical to the truncated expansion of Eqs. (15) and (16), evaluated in a coefficient-free form.

We place Chebyshev–Gauss–Lobatto nodes in A (N_A+1 nodes, with A=1 at spatial infinity and A=0 on the inner axis) and Gauss–Legendre nodes in B (N_B interior nodes that avoid the polar edges $B = \pm 1$ and, being a full nodal grid, carry both z-parities). Boundary conditions are imposed by row-replacement: Dirichlet $u = 0$ at the $A = 1$ (infinity) edge and Neumann $\partial_A u = 0$ on the $A = 0$ inner axis; on that inner axis only the $m = 0$ mode is regular, so we impose $u_m = 0$ there for $m \neq 0$, consistent with the $m = 0$ Neumann row. The Laplacian's rows span a huge dynamic range: the $(1-A^2)^4$ factor is tiny near infinity and the $1/A$ factor large near the inner axis, giving a raw condition number $\sim 10^{13}$–10^{15}. Using row equilibration, i.e. scaling each row by $1/\max_j |M_{ij}|$, an exact rescaling of the equations that leaves the solution unchanged, collapses the condition number to $\sim 10^4$ and takes the solve to machine precision.

B. Newton iteration with an analytic Jacobian

We solve Eq. (3) for the nodal field $U_{ijk} = u(A_i, B_j, \phi_k)$ using Newton iterations. Starting from an initial iterate $u^{(0)}$, each step solves the linearized system

$$J\big(u^{(k)}\big)\, \delta u^{(k)} \;=\; -\,R\big(u^{(k)}\big), \tag{17}$$

for the increment $\delta u^{(k)}$ and updates $u^{(k+1)} = u^{(k)} + \delta u^{(k)}$. The residual and its exact Jacobian $\partial R/\partial U$ are

$$R(u) \;\equiv\; M_0\, u \;+\; \tfrac{1}{8}\,(\psi_{\rm BL} + u)^{-7}\,\hat{A}^2\big|_{\rm int}, \tag{18}$$

$$J(u) \;\equiv\; M_0 \;-\; \tfrac{7}{8}\,{\rm diag}\big[(\psi_{\rm BL} + u)^{-8}\,\hat{A}^2\big]_{\rm int}, \tag{19}$$

where M_0 denotes the discretized Laplacian of Eq. (14), including the row-replaced boundary conditions, and the subscript "int" indicates that the nonlinear source is applied only on the interior rows. Because the Bowen–York source couples the Fourier modes whenever the data are non-axisymmetric, we evaluate the nonlinear term pseudo-spectrally on the ϕ collocation grid.

The structure of the Jacobian dictates the organization of the solver. The linear operator M_0 is block-diagonal in the azimuthal Fourier mode m ($\partial_\phi^2 \to -m^2$), so each mode reduces to an independent two-dimensional (A, B) solve that is factored once by its row-equilibrated LU decomposition. The nonlinear contribution breaks this decoupling. Its Jacobian,

$$D_{\rm nl}(A, B, \phi) \;\equiv\; -\tfrac{7}{8}(\psi_{\rm BL} + u)^{-8}\hat{A}^2, \tag{20}$$

depends on ϕ for non-axisymmetric data, and multiplication by a ϕ-dependent function corresponds to convolution in Fourier space. Consequently, the nonlinear term couples the azimuthal modes, making the full Jacobian dense across m.

We therefore solve the coupled Newton system with the full Jacobian Eq. (19) in matrix-free form, requiring only the matrix–vector product $J\,\delta u$. It is assembled exactly as the residual: the linear per-mode blocks M_0 plus the nonlinear coupling computed as ${\rm rfft}_\phi[D_{\rm nl}\,\delta u]$, i.e. a pointwise multiplication in ϕ followed by an FFT. The resulting linear system Eq. (17) is solved with GMRES, preconditioned by the block-diagonal linear operator. Since this preconditioner differs from the full Jacobian only through the omission of the Fourier-mode coupling, a few GMRES iterations suffice, while the outer Newton iteration retains quadratic convergence until the row-equilibrated residual stagnates, at which point the best iterate is returned. This Newton–Krylov solve is also the operation invoked by the certified polish of Sec. IV C.

Finally, we specify the norm used to monitor convergence. Because the prolate operator's rows span the wide dynamic range of Sec. III A, the *raw* nodal residual $\|R\|_\infty$ is dominated by the badly-scaled rows near the inner axis: it floors well above machine precision and even *rises* with resolution. This is a scaling artifact, not a failure to satisfy the constraint. Row equilibration removes it: scaling each row by $s_i^{(m)} \equiv \max_j \big|(M_0^{(m)})_{ij}\big|$ leaves the solution unchanged, so the honest measure is the *row-equilibrated* residual $\max_m \max_i \big|\hat{R}_i^{(m)}/s_i^{(m)}\big|$, and it is this well-conditioned norm that the Newton iteration drives to zero. This residual is nonetheless a *solver-internal* diagnostic; the *physical* constraint violation of the emitted data, computed by a generic finite-difference monitor on an evolution grid, with no reference to this norm, is verified independently in Appendix A 2.

IV. PARAMETRIC SOLVER

So far we have described how to obtain one constraint-satisfying datum. The contribution of this paper is to obtain the datum *as a function of* the physical parameters, without a full elliptic solve at query time.

A. The parametric problem

We consider the parameter vector $\theta \equiv (q, b, \boldsymbol{\chi}^A, \boldsymbol{\chi}^B)$ at fixed total bare mass $M \equiv m_A + m_B$ (set to 1), with the spins carried as the Kerr-like *dimensionless* spins $\boldsymbol{\chi}^X \equiv \boldsymbol{S}^X/m_X^2$, each a Cartesian 3-vector referred to the bare puncture mass m_X. The per-puncture momenta are not interpolation axes but are determined by the quasi-circularity condition of Sec. V A. The map to a slice is algebraic: $m_A = Mq/(1+q)$, $m_B = M/(1+q)$, and the Bowen–York spin that enters the analytic source is $\boldsymbol{S}^X = \boldsymbol{\chi}^X m_X^2$. The interpolation axes are q, b, and the individual spin *components* χ_i^X, i.e. up to eight scalar axes for the full two-spin family.

Sampling the spins in χ_i^X keeps the parameter domain rectangular, because the physical Kerr bound $|\chi_i^X| \le 1$ is independent of the component masses. The spin axes are therefore fixed independent of q and symmetric about zero. Each interpolation axis is finally mapped affinely onto $[-1, 1]$, the natural domain of the Chebyshev basis.

B. Value (Lagrange) collocation

Because the ABT coordinate system and collocation grid are identical at every parameter node, all solutions are represented on the same discrete function space. Each nodal value therefore corresponds to the same physical location in the compactified coordinate domain throughout parameter space. This fixed representation makes tensor-product interpolation over the parameters well-defined: the interpolant acts directly on corresponding nodal values, producing an interpolated field on the same spectral grid. The resulting parametric model inherits the smoothness of the underlying family of solutions and can be evaluated at arbitrary parameter values before being interpolated to arbitrary spatial coordinates using the coefficient-free spectral interpolation described in Sec. III A. The parametric model is then,

$$\tilde{u}(\boldsymbol{x};\theta) \;=\; \sum_{i_1=0}^{Q_1}\cdots\sum_{i_d=0}^{Q_d} U_{i_1\cdots i_d}(\boldsymbol{x}) \prod_{k=1}^{d} L_{i_k}(\theta_k), \tag{21}$$

where $U_{i_1\cdots i_d}$ is the high-fidelity solve stored at the parameter node $(\theta_{1,i_1}, \ldots, \theta_{d,i_d})$ and L_{i_k} is the Lagrange cardinal function associated with the k-th parameter. Equation (21) defines the method: initial data at a new parameter value are obtained as a parameter-independent linear combination of the stored high-fidelity solutions.

We evaluate the cardinal functions in the numerically stable barycentric form [16],

$$L_{i_k}(\theta_k) \;\equiv\; \frac{\lambda_{i_k}/\big(\theta_k - \theta_{k,i_k}\big)}{\displaystyle\sum_{j_k=0}^{Q_k} \lambda_{j_k}/\big(\theta_k - \theta_{k,j_k}\big)}, \tag{22}$$

where $\lambda_{i_k} = (-1)^{i_k}\delta_{i_k}$, with $\delta_{i_k} = 1/2$ at the endpoints and unity otherwise, so that $L_{i_k}(\theta_{k,j_k}) = \delta_{i_k j_k}$. The nodes θ_{k,i_k} are the affine images of the Chebyshev–Gauss–Lobatto points on $[\theta_k^{\min}, \theta_k^{\max}]$.

The tensor-product interpolant is evaluated by contracting one parameter axis at a time, requiring $O(\prod_k (Q_k+1))$ operations per query. More important than the per-query cost is that the interpolant provides an accurate initial guess throughout parameter space. Consequently, the certified polish of Sec. IV C reaches tolerance in a median of two to three Newton iterations, against four to five from a cold start. Because the parameter-to-solution map is analytic, the interpolation error decays exponentially in the number of parameter nodes, at a rate fixed by the nearest singularity in the complex parameter plane. This exponential convergence is demonstrated in Sec. V C.

C. Certified refinement

The parametric model of Eq. (21) supplies only a *guess*. We certify it with a few Newton steps of the genuine elliptic solver of Sec. III B. Taking the barycentric prediction as the initial iterate, $u^{(0)} = \mathcal{I}[u](\theta)$, we then apply the Newton iteration of Eq. (17), with the residual R and analytic Jacobian J of Eqs. (18,19). Because the guess is already close, a few steps drive the constraint residual to $\|R\|_\infty \le 10^{-10}$ at *any* queried parameter, *independent* of the interpolation error. This is the certification guarantee; Sec. V D makes it quantitative for the shipped models.

Two quite different quantities measure how good a guess $u^{(0)}$ is: its *field* error $\|u^{(0)} - u_{\rm true}\|_2/\|u_{\rm true}\|_2$, the ordinary distance between fields, which the interpolation accuracy controls; and the *constraint residual* $\|R(u^{(0)})\|_\infty$, which is what the Newton iteration (17) actually reduces. These are nearly *decoupled*. The residual operator differentiates the field (it contains the Laplacian, at scale $\propto b^{-2}$), so it amplifies precisely the small-amplitude, high-derivative content that the field norm barely registers: a guess can be excellent in field-norm yet violate the constraint by orders of magnitude more. Concretely, projecting the *exact* solution onto a smooth low-dimensional subspace leaves a field error $\sim 10^{-4}$ but a residual $\|R\|_\infty \sim \mathcal{O}(1)$, and it certifies in no fewer Newton steps than the crude barycentric guess. The step count is therefore set by $\|R(u^{(0)})\|_\infty$, not by field accuracy, with two consequences. First, refining the interpolant's field accuracy is a weak lever on the online cost, so the certified polish, which drives $\|R\|_\infty \to 0$ regardless of the guess, is the correct instrument. Second, one may discard the high-derivative content of the fields with no penalty, since the polish reconstructs it; this enables the reduced-basis re-encoding of Sec. IV H possible.

D. Differentiability

The parametric solver is differentiable with respect to the physical parameters: for any queried θ it exposes the sensitivities $\partial U/\partial\theta$ of the emitted field and, through any observable F constructed from it, the sensitivity $\partial F/\partial\theta$. That an interpolant is differentiable is unremarkable; the content of the claim is that these are derivatives of the *certified* initial data. The challenge is therefore not differentiating the interpolant, but preserving differentiability through the projection back onto the Einstein constraint manifold. Implemented in JAX [21] as a branchless barycentric map, the parametric model admits forward-mode automatic differentiation, which returns $\partial U/\partial\theta$ and $\partial F/\partial\theta$ directly, without numerical approximation.

At each node the tangent of the certified solution is available directly. Differentiating the vanishing residual identity $R(U(\theta),\theta) \equiv 0$ once gives the first-order forward-sensitivity equation

$$J\,\frac{\partial U}{\partial\theta_k} \;=\; -\,\frac{\partial R}{\partial\theta_k}, \tag{23}$$

with the same Jacobian $J = \partial R/\partial U$ (Eq. 19) that the certified solve has already factored, so each tangent costs one back-solve and no new factorization. What each axis needs is its own right-hand side, and on every axis of the box that derivative is closed-form. On the interior rows $\partial R/\partial\theta_k = \frac{1}{8}\big[(\psi_{\rm BL}+u)^{-7}\,\partial\hat{A}^2/\partial\theta_k - 7(\psi_{\rm BL}+u)^{-8}\,(\partial\psi_{\rm BL}/\partial\theta_k)\,\hat{A}^2\big]$, and it vanishes on the row-replaced boundaries. Because each Bowen–York tensor is linear in its own $\boldsymbol{P}_X$ or $\boldsymbol{S}^X$, every $\partial\hat{A}^2/\partial\theta_k$ is that same tensor re-evaluated with the derivative vector rather than a finite difference of the source. The spins enter only through $\hat{A}^2$, via $S^X = \chi^X m_X^2$; the mass ratio through the bare-mass map, $\partial\psi_{\rm BL}/\partial q = \sum_X (\mathrm{d}m_X/\mathrm{d}q)/2r_X$; and the separation through the fixed-node scale laws $\partial\psi_{\rm BL}/\partial b = -(\psi_{\rm BL}-1)/b$ and $\partial\hat{A}^{ij}/\partial b = -(2/b)\hat{A}^{ij}_P - (3/b)\hat{A}^{ij}_S$, together with the geometry term $-(2/b)\Delta u$ carried by the b^{-2} scaling of the prolate Laplacian, Eq. (14). An orbiting family adds one term on b, q and the aligned spins, since those axes also move the momenta through the quasi-circularity condition of Sec. V A: differentiating that closed-form prescription gives $\mathrm{d}\boldsymbol{P}_X/\mathrm{d}\theta_k$, whose momentum tensor enters $\partial\hat{A}^2/\partial\theta_k$ alongside the terms above. Nothing in the tangent is finite-differenced.

Two references verify the exposed sensitivities, and they play distinct roles (Appendix B). Second-order central finite differences of the same differentiable parametric model confirm the automatic differentiation itself. The load-bearing check is against the implicit-function-theorem tangent of Eq. (23), computed independently from the certified solve at the same parameter point: this is what establishes that the exposed gradient is the sensitivity of the certified datum rather than of the interpolant. Equation (23) is itself checked against finite differences of the certified three-dimensional solve. The remaining agreement is limited by the interpolation accuracy of the derivative, which converges exponentially in the parameter order just as the field interpolation does, and is correspondingly weakest along the slowly converging b and q axes.

Certification and differentiability therefore coexist rather than compete: every returned datum satisfies the constraints to tolerance, and its exact dependence on the physical parameters is available at once. This is what separates these sensitivities from those of a bare interpolant or a neural-network surrogate, and what makes the gradient-based applications of Sec. VI possible.

E. Gradient-enhanced (Hermite) collocation

Every certified solution carries verified parameter tangents, produced during the offline build (Sec. IV F) for the cost of a single back-solve against the node's already-factored Jacobian (Sec. IV D). This derivative information has therefore already been paid for, and to interpolate only the stored field values U_i of Eq. (21) would discard it. Gradient-enhanced (Hermite) collocation reincorporates it into the parametric model, matching values *and* parameter-derivatives at each node.

The gain is worth harvesting only where interpolation converges slowly. The fast parameter directions already converge spectrally, so matching tangents helps most in the slow directions near the Bernstein walls quantified in Sec. V C. We therefore enhance only a subset $\mathcal{E}$ of the axes, and leave every other axis with its plain Lagrange cardinal L_{i_k} of Eq. (22). Along an enhanced axis $k \in \mathcal{E}$ that cardinal is replaced by the Hermite pair $(H_{i_k}, \hat{H}_{i_k})$, and when two axes are enhanced together consistency requires matching their mixed cross-derivative as well, giving the genuine tensor-product Hermite interpolant on the enhanced subspace. For the two enhanced axes used throughout this work, let i_A and i_B denote the two indices i_k with $k \in \mathcal{E}$, write $U \equiv U_{i_1\cdots i_d}(\boldsymbol{x})$ for the certified solve stored at the node, and suppress the cardinal arguments, $H_{i_A} \equiv H_{i_A}(\chi_y^A)$ and $\hat{H}_{i_B} \equiv \hat{H}_{i_B}(\chi_y^B)$. The parametric model of Eq. (21) then generalizes to

$$\tilde{u}(\boldsymbol{x};\theta) \;=\; \sum_{i_1=0}^{Q_1}\cdots\sum_{i_d=0}^{Q_d}\left(\prod_{k\notin\mathcal{E}} L_{i_k}(\theta_k)\right)\Big[\,H_{i_A}H_{i_B}\;U \;+\; \hat{H}_{i_A}H_{i_B}\;\partial_{\chi_y^A}U \;+\; H_{i_A}\hat{H}_{i_B}\;\partial_{\chi_y^B}U \;+\; \hat{H}_{i_A}\hat{H}_{i_B}\;\partial_{\chi_y^A}\partial_{\chi_y^B}U\Big], \quad (24)$$

whose four terms are the value, the two certified first tangents, and the single bilinear cross tangent stored at the node (Sec. IV D). Each non-enhanced axis carries only its Lagrange cardinal, so restoring L on the two enhanced axes and dropping the tangents recovers the value interpolant of Eq. (21). The osculatory Hermite cardinals are built from the same barycentric Lagrange cardinal L_i of Eq. (22),

$$H_{i_k}(\theta) \;\equiv\; \big[\,1 - 2\,L'_{i_k}(\theta_{i_k})\,(\theta-\theta_{i_k})\,\big]\,L_{i_k}(\theta)^2, \qquad (25)$$

$$\hat{H}_{i_k}(\theta) \;\equiv\; (\theta-\theta_{i_k})\,L_{i_k}(\theta)^2, \qquad (26)$$

with $L'_i(\theta_i) = \sum_{j\neq i}(\theta_i - \theta_j)^{-1}$ the diagonal of the barycentric differentiation matrix. They satisfy $H_{i_k}(\theta_{j_k}) = \delta_{i_k j_k}$, $H'_{i_k}(\theta_{j_k}) = 0$, $\hat{H}_{i_k}(\theta_{j_k}) = 0$, and $\hat{H}'_{i_k}(\theta_{j_k}) = \delta_{i_k j_k}$, so the interpolant reproduces the certified value *and* tangents at every node by construction, and the exposed gradient at the nodes equals the certified-solve sensitivity exactly rather than the derivative of a value interpolant. Because the parameter-to-solution map is analytic, matching the tangent raises the per-node convergence order, so the same warm-start quality is reached with fewer expensive elliptic solves.

For both shipped models the enhanced set is the pair of aligned spin components, $\mathcal{E} = \{\chi_y^A, \chi_y^B\}$, identified as the slowest axes in Sec. V B. The remaining axes carry plain Lagrange cardinals: b and q in the four-dimensional aligned-spin model $\theta = (b, q, \chi_y^A, \chi_y^B)$, and those two together with the four in-plane components $\chi_x^A, \chi_z^A, \chi_x^B, \chi_z^B$ in the eight-dimensional general-spin model $\theta = (b, q, \boldsymbol{\chi}^A, \boldsymbol{\chi}^B)$. The bracket of Eq. (24) is therefore identical in the two models; only the number of Lagrange factors differs.

Restricting enhancement to $\mathcal{E}$ also keeps the stored-derivative count small: a full tensor-product Hermite over *all* d axes would carry mixed partials up to order d (2^d fields per node), whereas Eq. (24) carries only the $2^{|\mathcal{E}|} = 4$ derivatives internal to the enhanced subspace. The cross term is as cheap as the first tangents: differentiating the vanishing residual identity $R(U(\theta),\theta) \equiv 0$ twice gives the second-order forward-sensitivity equation

$$J\,\frac{\partial^2 U}{\partial\theta_k\,\partial\theta_{k'}} \;=\; -\Big[\,\frac{\partial^2 R}{\partial\theta_k\,\partial\theta_{k'}} \;+\; \frac{\partial^2 R}{\partial\theta_k\,\partial U}\,\frac{\partial U}{\partial\theta_{k'}} \;+\; \frac{\partial^2 R}{\partial\theta_{k'}\,\partial U}\,\frac{\partial U}{\partial\theta_k} \;+\; \frac{\partial^2 R}{\partial U^2}\Big[\frac{\partial U}{\partial\theta_k},\frac{\partial U}{\partial\theta_{k'}}\Big]\Big], \qquad (27)$$

with the *same* factored Jacobian $J = \partial R/\partial U$ as the first tangent (Eq. 19), i.e. one further back-solve and no new factorization. Because the spins enter the residual only through $\hat{A}^2$ (Sec. IV D), every term on the right is a closed-form, node-diagonal function of the source and its first two spin-derivatives, so no finite differencing is needed. Supplying this cross term closes the off-node cross-curvature between the two enhanced axes that a gradient-only construction, carrying the two first tangents but not their cross, leaves approximated. Completing it is not a refinement but the reason the enhanced pair works at all: tangents alone on two axes degrade the joint interpolant, and it is the cross that recovers them and carries the model past the value-only baseline (Sec. V E). The model assembled this way is the full-bilinear gradient-enhanced family measured in Sec. V D. Enlarging $\mathcal{E}$ beyond two axes would call for the higher mixed partials of the correspondingly larger tensor product, and we do not pursue it: Sec. V E finds that a larger enhanced set degrades the model rather than improving it.

The gradient-enhanced model is thus more accurate at no additional expensive cost: it reuses derivative information already produced by the offline build, introduces no new elliptic solves, and concentrates its accuracy gain along the slowly converging spin directions where interpolation is hardest. What it does cost is storage, since the tangents and their cross are kept alongside the value at every node; Sec. V G weighs the two families against stored memory and gives the reduced-basis re-encoding that removes most of it. Certification is unaffected: the enhanced model remains only a warm start, refined to tolerance by the certified Newton step of Sec. IV C.

F. Building the grid: warm-started continuation

The construction of the parametric model is independent of the underlying elliptic solver. It requires only a routine `solve`(θ, guess) that computes the solution at a parameter point θ from an initial guess and, optionally, the parameter tangent $dU/d\theta$. The interpolation nodes are populated by traversing the tensor grid in a boustrophedon (reflected-serpentine) order, so that successive nodes only differ by a single parameter increment. Each Newton solve can therefore warm-start from the converged solution at a neighbouring parameter point, while only the first solve requires a cold start.

G. Sparse grids for higher dimensions

The full tensor-product interpolant of Eq. (21) requires one expensive elliptic solve per parameter node, so its offline cost $\prod_k (Q_k{+}1)$ grows exponentially with the number of parameters d. This scaling is unnecessary because the parameter-to-solution map is analytic. Analytic dependence makes the interpolation error decay geometrically with the per-axis resolution (Sec. IV B), so refining every direction to high order is wasteful: the accuracy is dominated by a few finely resolved parameters rather than by resolving all of them simultaneously. A *Smolyak sparse grid* [13, 24] retains only these dominant tensor-product contributions.

Index each axis by a resolution *level* l_k, and let I_{l_k} be the one-dimensional interpolation operator on the $m(l_k)$ nodes of that level, with $m(0) = 1$ and $m(l) = 2^l + 1$ Clenshaw–Curtis doubling nodes[1]. These node sets are *nested*, and the levels telescope through the hierarchical surpluses $\Delta_l \equiv I_l - I_{l-1}$ (with $\Delta_0 \equiv I_0$), so that $I_L = \sum_{l\le L} \Delta_l$. For an analytic map the surpluses decay geometrically in l, through the same Bernstein-ellipse mechanism that governs the interpolation error (Sec. V C).

In d dimensions the interpolant telescopes into a sum of mixed surpluses over all levels, $\bigotimes_k I_L = \sum_{\boldsymbol{l}} \bigotimes_k \Delta_{l_k}$, recovering the $O(Q^d)$ cost. For an analytic map these mixed contributions decay geometrically in the total level $|\boldsymbol{l}|_1 = \sum_k l_k$, so resolving fine structure in several parameters at once is exponentially redundant. Retaining the *simplex* $|\boldsymbol{l}|_1 \le \ell$ captures nearly all of the interpolation accuracy and discards only exponentially small contributions. The retained surpluses recombine into the Smolyak combination technique,

$$\mathcal{I}_{\text{sparse}}[u] = \sum_{|\boldsymbol{l}|_1 \le \ell} c_{\boldsymbol{l}} \bigotimes_{k=1}^{d} I_{l_k}[u], \tag{28}$$

with the integer coefficients

$$c_{\boldsymbol{l}} = (-1)^{\ell - |\boldsymbol{l}|_1} \binom{d-1}{\ell - |\boldsymbol{l}|_1}, \tag{29}$$

nonzero only for $\ell - d + 1 \le |\boldsymbol{l}|_1 \le \ell$. Because the one-dimensional nodes are nested, the union of these subgrids is the sparse grid, and every node is solved only once.

The sparse grid replaces the exponential offline solve count with one that grows as $O(m\,(\log m)^{d-1})$ nodes versus $O(m^d)$, i.e. polynomially in the per-axis resolution and near-linearly in the dimension, while retaining nearly the full tensor-product accuracy. This scalability makes the higher-dimensional general-spin family tractable, where the full tensor grid is not, and the saving in offline solve count is quantified in Sec. V D.

Sparse grids alter only the offline sampling strategy: each subgrid interpolant in Eq. (28) is itself an instance of the parametric model of Eq. (21), so the online interpolation, the certified Newton refinement of Sec. IV C, and the differentiability of Sec. IV D carry over unchanged. When the parameters differ in difficulty, as the slowly converging spin axes of Sec. V B do, the isotropic simplex can be replaced by a *dimension-adaptive* downward-closed index set [13] grown greedily along the axes whose surpluses remain large, a refinement we leave to future work. In every case the sparse grid reduces the number of expensive offline solves without weakening the certification guarantee or the exact differentiability that define the method.

[1] The Clenshaw–Curtis nodes $\cos(\pi j/2^l)$, $j = 0, \ldots, 2^l$, are the extrema of the Chebyshev polynomials; taking $2^l{+}1$ of them (and thus doubling the number of subintervals each level) makes level l a subset of level $l{+}1$, i.e. nested, since $\cos(\pi j/2^l) = \cos(\pi(2j)/2^{l+1})$, such that the even nodes of level $l{+}1$ are equal to the nodes of level l.

H. Reduced-basis re-encoding

Because the model *is* its stored nodal solves (Eq. 21), its memory grows linearly in the node count, which becomes a liability in high dimension. However, since the parameter-to-field map is smoothly low rank, the corpus admits a lossless reduced-basis re-encoding.

We compress by *proper orthogonal decomposition* (POD, equivalent to principal-component analysis) [25, 26]. Stacking the N stored fields as rows of a matrix and subtracting their mean $\bar{u}$, a singular-value decomposition yields orthonormal *spatial* modes $\Phi = [\phi_1, \phi_2, \ldots]$ ordered by singular value σ_k; each stored field can then be written as $u = \bar{u} + \Phi\, c$ with coefficients $c = \Phi^\top (u - \bar{u})$. The singular values decay geometrically, because the map is analytic in θ, the same analyticity that governs the interpolation error (Sec. V C). Keeping the leading r modes, we store Φ together with the length-r coefficient vector at each node, i.e. r numbers per node in place of the full field, and, at query time, interpolate the *coefficients* with the identical barycentric/Smolyak machinery of Eq. (21) and decode $u(\theta) = \bar{u} + \Phi\, c(\theta)$.

Two properties make this a re-encoding rather than a lossy approximation. First, interpolation is linear, so interpolating the projected coefficients is *identical* to projecting the full-field interpolant onto the r leading modes: the compressed model is exactly the rank-r truncation of Eq. (21), differing from it only by a controllable truncation tail. Second, and this is why even that tail is harmless, the certified polish of Sec. IV C re-solves the constraint from the decoded guess, reconstructing precisely the high-derivative content the truncation discards; by the residual/field-accuracy decoupling above, the truncation changes the warm start's field error but not its Newton-step count. The differentiable evaluation of Sec. IV D also survives: the parameter Jacobian of the compressed model is the full Jacobian projected onto Φ, which loses a negligible fraction of the gradient. The decomposition is computed once from the existing corpus, with no new solves; Sec. V G reports the compression

TABLE I. **Parameter box and build configuration** shared by the two quasi-circular models (Sec. V). Both models sample the same edges on the same spatial grid at the same level: the aligned-spin model varies only χ_y^A and χ_y^B, the general-spin model varies all six components. The momenta follow from the other parameters through the quasi-circularity condition (Sec. V A).

puncture separation $D = 2b$ $[M]$	[6, 20]
mass ratio $q = m_A/m_B$	[1, 3]
spin components χ_i^A, χ_i^B	[−0.9, 0.9]
total bare mass $M = m_A + m_B$	1
Smolyak level ℓ	5
spatial grid (N_A, N_B, N_ϕ)	$(44, 32, 8)$
gradient-enhanced axes $\mathcal{E}$	$\{\chi_y^A, \chi_y^B\}$

achieved on the quasi-circular demonstration models.

V. QUASI-CIRCULAR PARAMETRIC MODELS

We now assemble the two headline models, both *quasi-circular* orbiting initial data built by the identical machinery over the production box $b \in [3, 10]\, M$ (i.e. puncture separations $D = 2b \in [6, 20]\, M$), $q \in [1, 3]$, and dimensionless spins with every component in $[-0.9, 0.9]$, at fixed total mass $M = 1$: a four-dimensional *aligned-spin* model $\theta = (b, q, \chi_y^A, \chi_y^B)$ and the full eight-dimensional *general-spin* model $\theta = (b, q, \boldsymbol{\chi}^A, \boldsymbol{\chi}^B)$ (Sec. V F), whose in-plane spin axes are centred on zero so the aligned model nests, node-for-node, inside it. Table I collects the box and the build configuration the two models share, for reference.

The accuracy metric throughout is the *held-out* interpolation error: at a parameter point that is *not* a collocation node we compare the interpolant to a direct elliptic solve on the *same* frozen spatial grid ($N_A = 44$, $N_B = 32$, $N_\phi = 8$) and take the largest nodal difference between the two fields. Because both are represented on the same grid, the spatial discretization error cancels and what remains is the pure parameter-interpolation error. We report it in two ways. The *per-axis* error sweeps a single parameter through 21 off-node points with the other axes held at a random base point, and keeps the worst of the 21; repeating that sweep at 100 random base points turns it into a distribution, and isolates the convergence rate of the swept axis (Fig. 1, and Fig. 2 at fixed rather than random base points). The *joint* held-out error instead draws 1000 parameter points uniformly at random from the whole box, so that every axis is off-node at once; it is the error encountered at an arbitrary query, and it is governed by the slowest-converging axis (Fig. 3). Both are distributions over the sampled points, which we report as a median with its extremes.

A. The quasi-circular momentum condition

What makes the family *orbiting* rather than a free momentum scan is that the per-puncture momenta are not free parameters but a deterministic function of (b, q, spins), fixed by a post-Newtonian quasi-circularity condition: a closed-form tangential momentum together with a small radial component that vanishes in the circular limit, so every node is a physical orbiting configuration and no free momentum axis is carried. The tangential momentum lies along x, off the separation (z) axis, so the orbital angular momentum points along $+y$ and the data are non-axisymmetric. We fix the momenta by the standard puncture quasi-circularity procedure [27, 28]. Defining $M \equiv m_A + m_B$, $\mu \equiv m_A m_B/M$, $\nu \equiv \mu/M$, coordinate separation $D \equiv 2b$, and $x \equiv M/D$, the per-puncture *tangential* momentum is the non-spinning post-Newtonian series through 3PN [28],

$$\frac{P_t}{\mu} = x^{1/2} + 2\,x^{3/2} + \tfrac{1}{16}(42 - 43\nu)\,x^{5/2} + \tfrac{1}{128}\big(480 + (163\pi^2 - 4556)\nu + 104\nu^2\big)\,x^{7/2}. \tag{30}$$

Note that the leading term is the Newtonian value $P_t \to \mu\sqrt{M/D} = \mu\sqrt{M/2b}$. We add the leading (1.5PN) spin–orbit correction [29], which depends only on the spin projections $\chi_\parallel^X$ onto $\hat{\boldsymbol{L}}$; in-plane spin components first enter at higher PN order and are neglected. Converted to the convention used throughout this paper[2] ($q = m_A/m_B \geq 1$, A the larger hole; $\chi_\parallel^X \equiv (\boldsymbol{S}^X \cdot \hat{\boldsymbol{L}})/m_X^2$ the dimensionless spin projected on the orbital angular momentum, i.e. the aligned component), it reads

$$\frac{P_t^{\rm SO}}{\mu} = \frac{2}{3(1+q)^2}\big[q(4q+3)\,\chi_\parallel^A + (3q+4)\,\chi_\parallel^B\big]\,x^2. \tag{31}$$

The tangential momentum that the data actually carry is the sum of the two,

$$P_t^{\rm tot} = P_t + P_t^{\rm SO}, \tag{32}$$

which reduces to Eq. (30) whenever both spins are non-spinning or purely in-plane. The *radial* component is the leading radiation-reaction (Peters) momentum $P_r = \mu|\dot{D}|$ with the quadrupole inspiral rate $\dot{D} = -\frac{64}{5}\mu M^2/D^3$ [28, 30], i.e.

$$\frac{P_r}{\mu} = \frac{64}{5}\frac{\mu M^2}{D^3}, \tag{33}$$

so $P_r/P_t \propto \nu\,(M/D)^{5/2} \to 0$ as $b \to \infty$, yielding the circular limit. The per-puncture momenta are then

[2] Healy *et al.* [29] write this correction with $q = m_2/m_1 \leq 1$ (body 1 the larger hole); Eq. (31) is the algebraically identical form obtained by substituting $q \to 1/q$ and relabelling $(1, 2) \to (A, B)$. It is invariant under $q \to 1/q$ with $\chi_\parallel^A \leftrightarrow \chi_\parallel^B$, as it must be.

$\boldsymbol{P}_A = (+P_t^{\rm tot}, 0, -P_r)$, $\boldsymbol{P}_B = (-P_t^{\rm tot}, 0, +P_r)$, with zero net linear momentum and orbital angular momentum $2bP_t^{\rm tot}$ along $+y$; only the aligned (S_y) spin component enters Eq. (31), so in-plane spins leave the momenta unchanged at this order. The angular momentum this produces is validated against its closed form and against TwoPunctures, and the large-separation Newtonian limit of Eq. (30) is the cross-check that the two codes share a momentum and separation convention (Appendix A 1).

B. Convergence across the parameter space

Sweeping one parameter at a time, every axis converges geometrically (Fig. 1). The separation b and mass ratio q are the fastest; the spins are somewhat slower, but all eight rates lie within a narrow band, and the two holes' spin axes converge at comparable rates. Gradient enhancement (Sec. IV E) roughly doubles every rate for the price of one extra back-solve per node against the node's already-factored Jacobian. Because the joint held-out error is governed by the slowest of these axes, it is the spin directions that set the offline cost of the model (Fig. 3 and Sec. V D).

C. Analyticity limits of the parameter map

The convergence rate of a spectral parameter interpolant is set by the distance from the sampling interval to the nearest singularity of the parameter-to-solution map in the complex plane, through the *Bernstein-ellipse* convergence bound (Theorem 8.2 of Ref. [16], going back to [31]). We map the sampling interval $[\theta_{\rm min}, \theta_{\rm max}]$ of a parameter θ affinely onto $[-1, 1]$ by $\theta \mapsto \xi = (2\theta - \theta_{\rm max} - \theta_{\rm min})/(\theta_{\rm max} - \theta_{\rm min})$, and consider the *Bernstein ellipses* E_ϱ, which are defined as the ellipses in the complex ξ-plane with foci at ± 1 whose semi-major and semi-minor axes sum to $\varrho > 1$. If the parameter-to-solution map is analytic inside E_ϱ, its Q-node Chebyshev interpolant converges geometrically, with error $\sim \varrho^{-Q}$, i.e. a rate of $\log_{10} \varrho$ decades per node. The operative ϱ is fixed by the *largest* ellipse free of singularities, so the nearest singularity of the map lies *on* E_ϱ. A singularity at the mapped point ξ_* sits on the ellipse of parameter $\varrho = |\xi_* + \sqrt{(\xi_*)^2 - 1}|$, which for a real singularity outside the interval reduces to $\varrho = |\xi_*| + \sqrt{(\xi_*)^2 - 1}$. Hence, a singularity in parameter space close to the sampled box $[\theta_{\rm min}, \theta_{\rm max}]$ will yield a ϱ just above unity and thus slow convergence, while a distant singularity will yield a large ϱ and fast convergence.

We read this estimate *backwards* to locate the wall. From the slope of the held-out error against the node count Q we measure the geometric convergence rate $r \equiv \log_{10} \varrho$, invert it to $|\xi_*| = \frac{1}{2}(\varrho + \varrho^{-1})$, and map ξ_* back through the affine map, taking the real root outside the sampled box on the physically singular side (small b toward merger for the separation; large q toward the test-mass limit for the mass ratio; large $|\chi|$ toward the extremal limit for the spins), to read off the inferred singularity location θ_*. Sweeping the sampling range then diagnoses the *character* of the wall. A genuine real branch point stays pinned at a fixed θ_* as the range changes. A complex-conjugate pair off the real axis has no real singularity at all, yet the one-parameter fit still returns a real θ_*, the real point lying on the same Bernstein ellipse as the true complex pair, which *marches* outward as the range grows. The legends of Fig. 2 report this inferred θ_* for each sampling range alongside the fitted rate; whether θ_* holds still or recedes as the range grows separates the two wall types below.

The quasi-circular family has three walls, one hard and two soft (Fig. 2). The separation and spin sweeps are run at equal mass with the remaining parameters at their symmetric values; the mass-ratio sweep, whose swept axis is itself the mass ratio, instead holds $b = 3.5\,M$ and both spins at zero.

The *merger* wall, in the separation b, is hard. As the sampling interval is pushed toward coincidence the rate degrades smoothly and tracks the Bernstein prediction for a singularity fixed at $b = 0$, the inferred b_* staying at the merger throughout. This resolves the concern (specific to the orbiting family) that the $b \to 0$ limit is *doubly* singular: the punctures coincide *and* the post-Newtonian tangential momentum $P_t \propto b^{-1/2}$ diverges, yet the extra branch does not *relocate* the wall off $b = 0$. Its only signature is a mild undershoot of the measured rate below the pure-coincidence prediction at the smallest $b_{\rm min}$, i.e. the momentum divergence makes the wall marginally *stronger*, not closer.

The *mass-ratio* wall is soft. Widening the sampled interval degrades the rate only mildly, from 0.62 to 0.52 decades per node between $q_{\rm max} = 3$ and $q_{\rm max} = 4$, while the inferred nearest real singularity recedes from $q_* = 4.19$ to 5.22, holding $q_*/q_{\rm max}$ near 1.3–1.4 throughout. A feature that retreats in proportion to the interval is a complex-conjugate pair rather than a real branch point, so the interval never approaches it and $q_{\rm max} = 3$ is not an accuracy limit: covering the asymmetric mass ratios out to $q = 4$ would cost some 15% of the rate. The production edge is therefore a scope choice, not a wall.

The *spin* wall, in χ, is by contrast soft. The measured rate degrades only mildly as the range grows. Here, the range was deliberately extended past the production box to locate the wall and the inferred nearest *real* singularity marches outward, staying beyond the sampled range throughout, which is the signature of a complex-conjugate pair off the real axis rather than of a real wall. That nearest feature is already super-extremal, well past the Cook–York horizon-spin ceiling [32], so spin is a benign axis with no sharp singularity near the physical box. What sets the convergence rate is therefore the *distance* of the nearest singularity and not its hardness: the family's only hard wall lies on one of its two fastest axes, while its slowest axes carry only soft, receding ones.

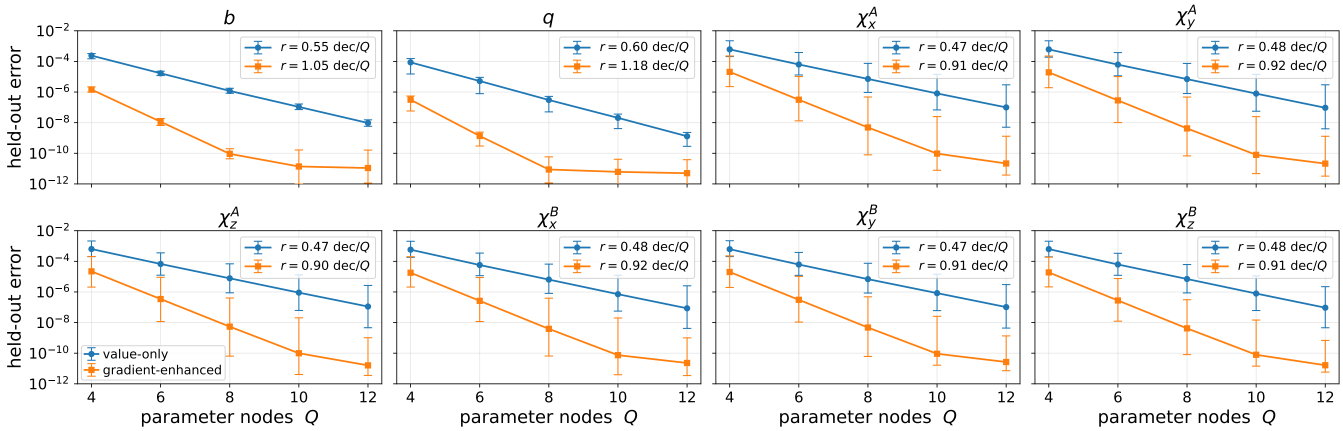

FIG. 1. **Per-axis convergence of the quasi-circular models.** Held-out interpolation error against the number of parameter nodes for each of the eight axes $(b, q, \boldsymbol{\chi}^A, \boldsymbol{\chi}^B)$; the first four are the four-dimensional aligned-spin model. The value-only model is compared with the gradient-enhanced one. A one-parameter sweep carries no mixed tangent, so the cross term of Eq. (24) does not enter here. Each plotted error is the worst over 21 off-node points along the swept axis; markers are medians of that worst case over 100 random base points for the non-swept axes, with whiskers spanning its best–worst range across those base points. Every axis decays geometrically and matching the tangent roughly doubles each rate.

D. Cost, memory, and certified residuals

The offline cost of a full tensor grid scales as $\prod_k (Q_k+1) = O(Q^d)$; a sparse (Smolyak) grid [13, 24] breaks this and is what makes both the four-dimensional and the eight-dimensional quasi-circular models (Sec. V F) tractable. Figure 3 is that cost curve: joint held-out accuracy against the number of elliptic solves that bought it, level by level. At isotropic Smolyak level $\ell = 5$ the four-dimensional model is 1105 solver nodes and the eight-dimensional one 15,713, and the nested lower levels $\ell = 1 \subset \cdots \subset 5$ are fully contained in each corpus, so no lower level costs additional solves. A full tensor grid at the four-dimensional sparse grid's finest level (33 nodes per axis) would be $\sim 10^3\times$ that node count, and one pushed to a *bare* held-out accuracy of 10^{-6}–10^{-9} two to four orders of magnitude more, an accuracy the certified construction never needs, because polish carries the correctness.

Evaluating the interpolant is cheap. A bare query is a single contraction over the stored nodal solves, with no Newton iteration and no factorization, and costs only a small fraction of one Newton–Krylov step of the elliptic solve, and less still after the reduced-basis re-encoding of Sec. V G. The warm start it provides is therefore essentially free relative to the solve it seeds. A certified query is that same warm-started solve, and its saving over a cold solve is the reduced step count: two to three polish steps rather than the four to five a cold start needs. The practical pattern is thus to bulk-generate initial data with bare queries and certify only the configurations one actually evolves. Because the model is nothing but its stored nodal solves, its memory is *linear* in the node count.

E. How many axes to enhance

Because the enhanced subspace is where the extra back-solves are spent, we tested whether $\mathcal{E}$ should be enlarged beyond the two aligned-spin axes. It should not. Enhancing a *single* axis always improves on the value interpolant, but enhancing several axes with tangents alone degrades it, and it does so whichever axes are chosen. Extending $\mathcal{E}$ to all four axes of the aligned-spin box at the same Smolyak level costs two orders of magnitude in held-out median relative to enhancing nothing at all, and completing the pairwise-bilinear crosses does not repair that. The two-axis case is the one that can be rescued: with tangents alone the aligned-spin pair is also worse than value-only, and it is the bilinear cross term of Eq. (24) that turns it into the model shipped in Fig. 3.

The damage belongs to the sparse-grid combination rather than to any one ingredient. We checked the reduced-basis compression, the cross layer, and the one-dimensional Hermite operators in isolation; swept one axis at a time, Hermite beats Lagrange on *every* axis, so the one-dimensional rules are sound. What correlates with the degradation is only the *number* of simultaneously enhanced axes.

The mechanism we expect is the following, although we have not established it directly. The Smolyak interpolant of Eq. (28) is a *signed* sum: neighbouring subgrid interpolants enter with opposite signs and cancel almost completely, so the retained result is a small difference of much larger terms, and its error is limited by the size of those terms and not by their difference. That size is set by the operator norms of the one-dimensional rules the term is built from, one factor per axis, so the norms *multiply* across the enhanced dimensions. The osculatory Hermite cardinals of Eqs. (25)–(26) are quadratic in

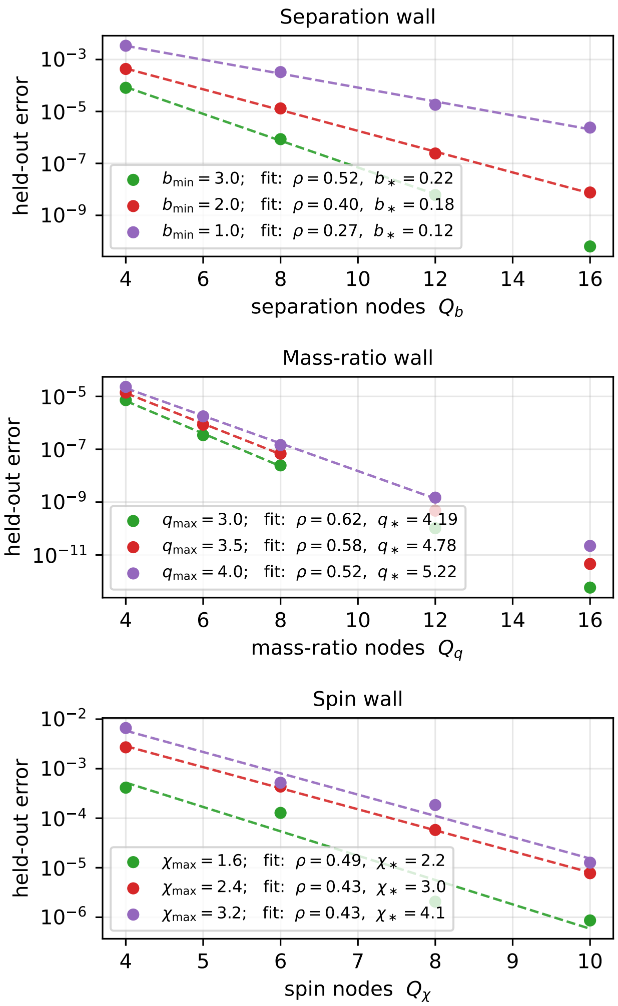

FIG. 2. **Analyticity walls of the quasi-circular family.** Held-out interpolation error against the number of nodes on the swept axis, for several sampling ranges, with the geometric fit $\varepsilon \approx A\,10^{-\rho Q}$ dashed over the fitted window, which ends where the error saturates on the float64 floor. Legends give, per range, the fitted rate ρ (decades/node) and the inferred nearest real singularity θ_*. *Top (separation b):* the rate degrades as the interval is pushed toward coincidence while b_* stays pinned at the merger, i.e. a hard, real wall. *Middle (mass ratio q):* the rate degrades only mildly and q_* recedes roughly in proportion to the interval, so the box never approaches it, i.e. a complex pair, hence soft. *Bottom (spin χ):* the rate degrades only mildly and χ_* marches outward, beyond the sampled range, i.e. again a complex-conjugate pair off the real axis, hence soft (Sec. V C).

the Lagrange cardinal and so carry a larger norm than it does, whereas what they buy, one additional order of accuracy per node, enters only once per axis. One enhanced axis is therefore a clear win; each further one multiplies a growing amplification against an additive gain, until the amplification wins.

The practical consequence is that $\mathcal{E}$ should be kept small and spent on the slowest axes, and that the shipped $\mathcal{E} = \{\chi^A_y, \chi^B_y\}$, completed to full bilinear, is at or near the optimum of the configurations we measured.

F. The eight-dimensional general-spin model

Promoting the two aligned spins to full vectors takes the model to eight dimensions, $\theta = (b, q, \boldsymbol{\chi}^A, \boldsymbol{\chi}^B)$, where the tensor grid's Q^d cost makes the Smolyak construction of Sec. IV G not merely advantageous but necessary. The method is unchanged: with the in-plane spin axes centred on zero, the four-dimensional aligned model nests, node-for-node, inside the eight-dimensional one, and the $\ell = 5$ grid is 15,713 solver nodes, reusing the entire aligned corpus and adding only the genuinely-new precessing nodes. Its in-plane spin axes converge like the aligned ones, the two holes at comparable rates (Fig. 1), and the in-plane spin wall is again soft, its nearest real singularity receding as the fit range grows, so precession introduces no new hard wall. At this dimension it is the polish, not the bare sparse interpolant, that carries the accuracy: the 1000-point certified-polish statistics for both $\ell = 5$ models are reported in Fig. 4. The gradient-enhanced corpus reduced-basis compresses by a factor of 30 with no loss of certification or differentiability (Sec. V G).

G. Reduced-basis compression of the model

The corpora are correspondingly large: the gradient-enhanced four-dimensional $\ell = 5$ corpus (values, the d node tangents, *and* the bilinear cross tangent) is 486 MiB and the eight-dimensional one 12.1 GiB. This is a distribution liability for an artifact meant to be hosted and cited, and it is entirely removable by the reduced-basis (POD) re-encoding of Sec. IV H: an SVD of the stored fields yields orthonormal spatial modes Φ, of which we keep the leading r and interpolate the length-r coefficient vector at each node in place of the full field. The singular values decay geometrically (the map is analytic in θ, as in Sec. V C): for the four-dimensional model the rank-r reconstruction error of the corpus falls below 10^{-6} by $r = 87$ modes, and because the value and its parameter tangents *share* one spatial basis, the enhanced corpus needs only some 7% more modes than the value-only one at the same tail. The rank at a fixed accuracy is set by the intrinsic dimension of the solution manifold, not by the sampling density: it is the counterpart to Sec. IV G along the orthogonal axis — the Smolyak level fixes the parameter resolution and the offline solve count, POD the spatial rank, and the two are independent.

Figure 5 shows the resulting memory–accuracy trade-off. At the shipped ranks the four-dimensional model shrinks by a factor of 49 (486 MiB to 10.0 MiB) and the eight-dimensional one by 30 (12.1 GiB to 420 MiB), and per-query evaluation, now a barycentric sum over r coefficients rather than the full field, drops to a few

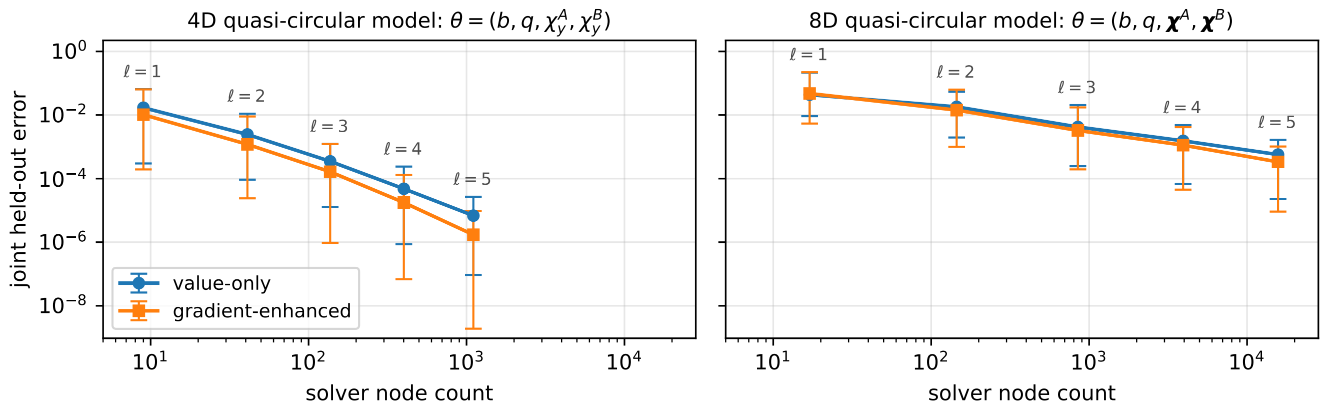

FIG. 3. **Joint held-out convergence.** Joint held-out error over 1000 random off-node points against solver node count (logarithmic, shared between panels), with the Smolyak level $\ell = 1$–5 labelled at each point, shown as the best, median, and worst of the distribution. *Left (four-dimensional model* the value-only sparse interpolant against the full-bilinear gradient-enhanced model enhanced on the two spin axes, which is more accurate at every level and reaches a $\sim 35\times$ deeper best-case floor at the same node count, from the node tangents alone. *Right (eight-dimensional model):* the same two curves.

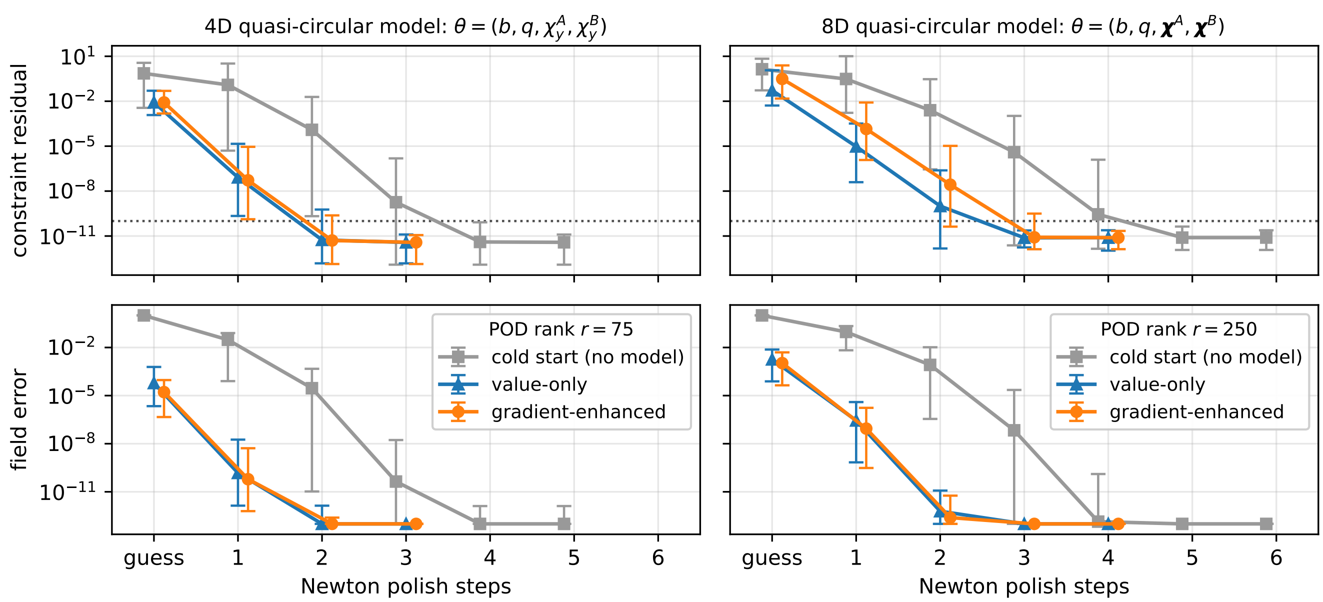

FIG. 4. **Certified refinement.** Constraint residual $\|R\|_\infty$ (top row) and relative field error $\|u - u_{\mathrm{true}}\|_2/\|u_{\mathrm{true}}\|_2$ (bottom row) over 1000 uniformly-random off-node points of the quasi-circular models (χ, $b \in [3, 10]\,M$, Smolyak $\ell = 5$; left column four-dimensional aligned-spin, right column eight-dimensional general-spin), before the polish ("guess") and after each Newton step. Each curve is the median with min–max whiskers over the 1000 points, the field error measured against the converged certified iterate. *Dotted:* the certification threshold 10^{-10}. Both value-only and gradient-enhanced models are POD-compressed to the same rank so that the two differ only in the gradient enhancement. Both warm starts certify *every* point to $\|R\|_\infty \leq 10^{-10}$ in at most three Newton steps in four dimensions and four in eight.

milliseconds. The certified constraint residual, the number of Newton steps needed to reach it (Sec. IV C), and the verified $\partial\mathrm{ID}/\partial\theta$ (Sec. IV D) are all preserved: the mode projection loses only the truncation-tail fraction of the gradient, and the compressed model reproduces the truncated interpolant to that tail and reloads as a standalone predictor. The decomposition is computed once from the existing corpus, in minutes and with no new solves.

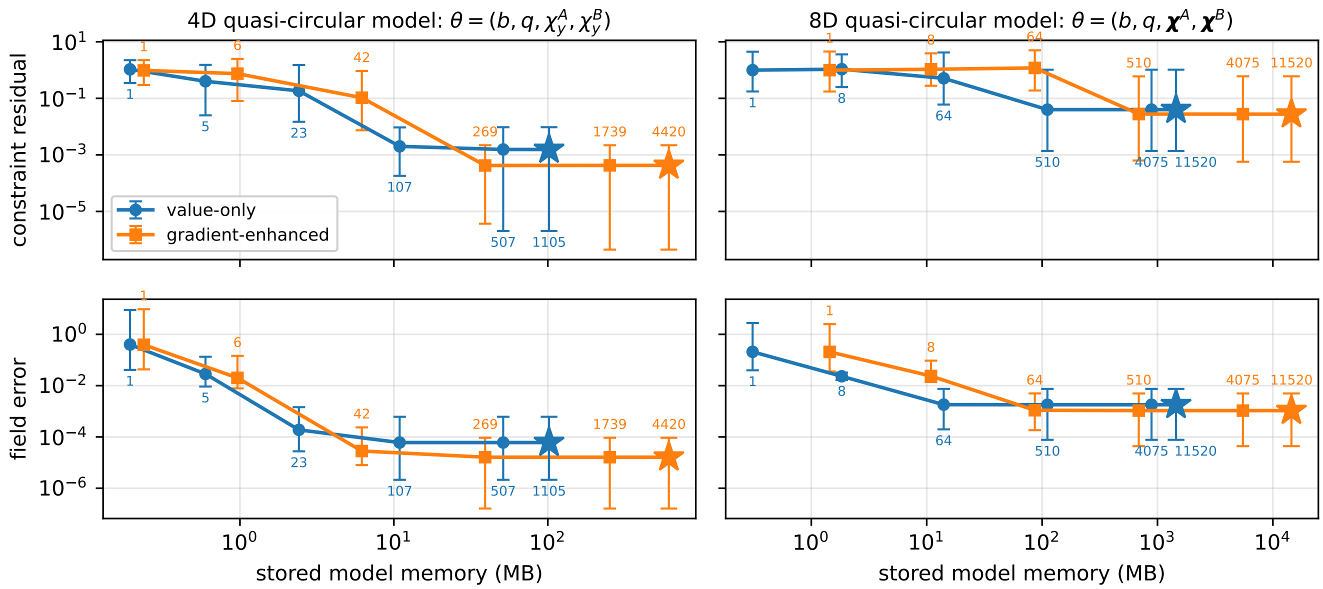

FIG. 5. **Reduced-basis (POD) compression versus stored memory.** Constraint residual $\|R\|_\infty$ (top row) and relative field error $\|u - u_{\rm true}\|_2/\|u_{\rm true}\|_2$ (bottom row), as the POD truncation rank r is swept. Median line with min–max whiskers over 1000 uniformly-random off-node query points; each point is labelled by r and $\star$ marks the full-rank model. Columns share the memory axis: *left,* the four-dimensional and *right,* the eight-dimensional quasi-circular model. Compressing raises the residual, but this warm-start quality is irrelevant to correctness: the certified polish (Sec. IV C) drives every query to $\|R\|_\infty \le 10^{-10}$ in the same Newton-step count regardless. The field error *saturates* at a far lower rank than the residual, so the redundancy discarded by compression is exactly the high-derivative content the residual amplifies and the field norm cannot see (the decoupling of Sec. IV C).

H. External validation

The underlying initial data reproduce the established TwoPunctures [23] solver to spectral accuracy on the most demanding, genuinely non-axisymmetric configurations, and they are evolution-ready: on an evolution grid their finite-difference constraint violation is indistinguishable from TwoPunctures-sourced data. The two codes discretize differently, so this is an independent cross-check of the implementation rather than a restatement of the certification, which is measured internally on our own grid (Sec. IV C). Appendix A gives the full comparison.

VI. APPLICATIONS

The two payoffs of the method are that it is differentiable and constraint-certified, and we demonstrate both in two applications: certified parameter targeting on a four-dimensional quasi-circular model, and eccentricity reduction on a purpose-built two-dimensional (b, P_t) model. Each targets a standard numerical-relativity initial-data workflow, and in each the honest cost metric is the number of *certified* elliptic solves.

A. Certified parameter targeting

The first workflow is parameter control [5]: find the free data that realize a chosen physical configuration. The standard tool is a black-box control loop, i.e. an outer Newton/Broyden iteration that adjusts the free data and performs a full constraint solve at every step to measure the resulting physical parameters. We target the ADM mass and angular momentum $(M_{\rm ADM}, J)$ by adjusting (b, q) at zero spin, over 100 random known-answer targets, and compare three strategies, all on the four-dimensional aligned-spin model of Sec. V. A *cold* black box solves the constraints from scratch at each control evaluation. A *warm* black box seeds each solve from the interpolant, so every constraint solve begins close to its answer. The *differentiable* method runs the entire Gauss–Newton control loop on the free parametric model using the analytic $\partial F/\partial\theta$, then certifies the result with a few Newton steps (i.e. no elliptic solve per step, and no finite-difference Jacobian).

Differentiability changes the cost class here (Fig. 6). The gradient method reaches every target in at most three certified solves, with a median of two, whereas both black-box variants take a target-dependent seven to fourteen, with a median of ten. Warm-starting cuts the per-solve cost but *not* the number of solves: the outer control iteration is unchanged, so a warm start alone never

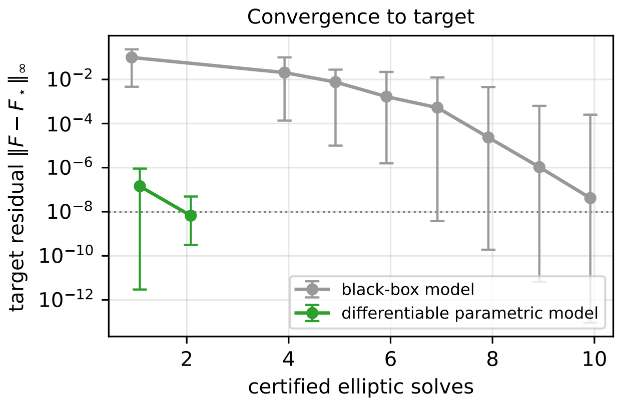

FIG. 6. **Certified parameter targeting.** Reaching a physical target $(M_{\rm ADM}, J)$ by adjusting (b, q), over 100 random known-answer targets, measured in *certified* elliptic solves. Each method is drawn as the across-target median residual at each solve count, with min–max whiskers. Running the control loop on the differentiable parametric model reaches tolerance in at most three certified solves for every target, with a median of two; the black-box loop takes a target-dependent seven to fourteen, with a median of ten. Measured on the four-dimensional aligned-spin model over the production box of Table I, at zero spin; every configuration emitted is certified to $\|R\|_\infty \leq 10^{-10}$.

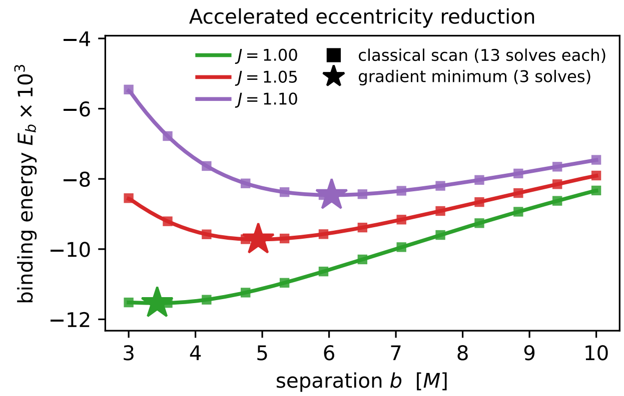

FIG. 7. **Accelerated eccentricity reduction.** At fixed angular momentum J the binding energy $E_b(b)$ has a minimum at the circular orbit; sweeping J the minimum shifts outward, tracing the circular-orbit sequence. For each J the differentiable $\partial E_b/\partial b$ locates that minimum in three certified solves, against the classical certified-solve scan's thirteen (squares). In the shallow small-J well the fixed scan mislocates the minimum, whereas the derivative still resolves it.

leaves the black box's complexity class. The distinction is structural. A black-box loop can drive the residual to zero only through function evaluations, and each one is a certified elliptic solve; a low-rank Jacobian update (Broyden's method, which sidesteps the d extra solves a finite-difference Jacobian would cost) lowers the iteration count but cannot remove the one solve per iteration. The analytic gradient removes those per-iteration solves outright, running the search entirely on the free parametric model and leaving only the last-mile certification, a saving that widens with the number of controlled parameters.

B. Eccentricity reduction

The second workflow is eccentricity control, the most common tuning step in preparing quasi-circular data. We use the Cook effective-potential method [33]: along a sequence of fixed angular momentum J (radial momentum $P_r = 0$, i.e. an apsis), the binding energy $E_b = M_{\rm ADM} - (M_A + M_B)$ has a minimum at the circular orbit $\partial E_b/\partial b|_J = 0$, and a mildly off-circular apsis at (b_0, J) has eccentricity $e = |b_0 - b'|/(b_0 + b')$ set by the second turning point b' across that minimum. Reducing eccentricity is driving $b_0 \to b_{\rm circ}$.

The classical method locates $b_{\rm circ}$ by *scanning* b at fixed J, i.e. one certified solve per point, and fitting the minimum. This workflow needs the tangential momentum P_t as a *free* axis (one scans b at fixed $J = 2bP_t$), which the quasi-circular model of Sec. V deliberately does not carry; we therefore build a dedicated small two-dimensional model in (b, P_t) (equal mass, no spin) by the identical machinery of Secs. IV–V. On it the differentiable binding energy exposes $\partial E_b/\partial b|_J$ analytically, so a Newton root-find locates the circular orbit on the free interpolant and certifies at the end. Sweeping J traces the circular-orbit sequence, whose minimum shifts outward with angular momentum (Fig. 7). The gradient locates each $b_{\rm circ}$ in three certified solves against the scan's thirteen, agreeing with it to 5×10^{-3} where the well is well resolved. It is moreover *more accurate* in the shallow, near-innermost-stable-orbit regime, where the fixed thirteen-point scan mislocates the minimum by half a mass and the derivative still resolves it: a black-box scan must sample ever more finely, at more certified solves, exactly where the physics is most delicate. The same differentiable E_b reads the orbital eccentricity off the turning points, from $e = 0$ at the circular orbit to $e \approx 0.23$ at the box edge, and so exposes $\partial e/\partial \theta$ for gradient-based eccentricity reduction.

Both workflows make the same point: the differentiable, certified map turns a black-box search (a control loop and an effective-potential scan) into a gradient problem solved on the free parametric model and certified in a handful of Newton steps, where the black box instead pays a full certified elliptic solve at every trial.

VII. DISCUSSION AND CONCLUSIONS

We have introduced a parametric solver for binary-black-hole puncture initial data that combines sparse interpolation with Newton refinement. The interpolation supplies an inexpensive parameter-dependent initial guess, while the subsequent elliptic solve certifies every returned datum to a prescribed constraint tolerance. Because the construction is implemented in JAX, the resulting parameter-to-solution map is also exactly differentiable with respect to the physical parameters. The four-dimensional aligned-spin and eight-dimensional general-spin models demonstrate that this construction remains practical across physically relevant binary-black-hole parameter spaces, while the parameter-targeting and eccentricity-reduction examples show how its derivatives can reduce the number of certified elliptic solves required in downstream workflows.

The two headline parametric models are genuinely three-dimensional and quasi-circular, spanning the aligned- and general-spin families (Sec. V). What remains is scope, not a barrier of principle. The most natural extension is *eccentricity*: relaxing the quasi-circularity condition to carry an independent radial momentum adds one axis on top of the orbiting family, by the same machinery — indeed we already build a small (b, P_t) eccentricity model this way (Sec. VI B). The parametric, certified, and differentiable machinery is *dimension-agnostic*, i.e. it sees only a $\texttt{solve}(\theta, \text{guess})$ routine and an optional tangent, so each extension is a change of solver and box, not of method.

One limitation is inherited from the free data themselves, not from the parametric method. Bowen–York, conformally flat data carry spurious ("junk") radiation that the early evolution must shed [34], and cannot represent near-extremal black holes: the *horizon* spin of a Bowen–York puncture saturates at $\chi \approx 0.93$ no matter how large the free-data spin parameter [32, 35], which is why nearly-extremal-spin simulations use conformally curved (e.g. superposed Kerr–Schild) initial data [36]. Within the moving-puncture mainstream ($\chi \lesssim 0.9$) the family built here is the standard input; and because the parametric layer is solver-agnostic (Sec. IV F), a conformally-curved solver could be swapped in under the same interpolation, certification, and compression machinery. We will explore this in future work.

Two accuracy caveats are worth stating explicitly. First, the spin-wall analysis is quantitative only within the physical range ($\chi_{\max} \lesssim 0.8$): pushing the sampling interval toward and past extremal spin mixes the genuine (complex, far) singularity with pre-asymptotic and spatial-floor effects, so the clean single-χ_* characterization is reported for the physical regime, not extrapolated beyond it. Second, the certified $\|R\|_\infty \leq 10^{-10}$ gate is verified at the production grid ($44 \times 32 \times 8$); at higher meridian resolution the whole-field equilibrated residual floors higher, because the stiff m^2/ρ^2 term amplifies roundoff in the azimuthal modes that carry no source content, while the physically populated modes stay three orders of magnitude below. The floor is therefore set by empty modes rather than by the physics, and the certification tolerance should be read at the grid it is quoted for.

Finally, beyond the immediate use cases, the method produces exactly the kind of object a broader surrogate or foundation-model programme for binary-black-hole spacetimes needs at its base: a fast, *exact* (constraint-certified), and *differentiable* initial-data generator across a physical parameter space. Such a generator can serve as a certified training source and a warm-start oracle for those models, supplying gradients $\partial \mathrm{ID}/\partial\theta$ for end-to-end parameter inference.

ACKNOWLEDGMENTS

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Appendix A: Validation against TwoPunctures

We validate the underlying initial-data solver against the established TwoPunctures code [23], using its standalone C port from the NRPy suite compiled against GSL. We compare the quasi-circular data over the whole production box (Appendix A 1), and then examine their behavior after interpolation onto an evolution grid (Appendix A 2). In the axisymmetric limit, the two independently discretized codes agree to approximately machine precision.

1. Quasi-circular data against TwoPunctures

We next validate the quasi-circular configurations used in Sec. V. Because our prolate coordinate system places the punctures on the z axis, whereas the standard TwoPunctures convention places them on the x axis, we first rotate both data sets to a common Cartesian frame. We also verified the mapping between the bare masses, half-separation, and Bowen–York momenta and spins used by the two codes. With these conventions fixed, both codes employ the same decomposition $\psi = \psi_{\mathrm{BL}} + u$, allowing a direct comparison of the conformal factor at common physical points.

As a first check, we consider the momentum prescription used to construct the quasi-circular data. In the equal-mass, nonspinning limit, the post-Newtonian expression of Sec. V approaches the Newtonian circular-orbit result at large separation. The corresponding Bowen–York momentum therefore provides a direct check that the separation and momentum conventions used in the two implementations are consistent.

We then compare the resulting initial data directly with TwoPunctures. So that the comparison does not rest on one arbitrary parameter point, we draw 100 configurations from the production box by a Latin hypercube and walk the same meridional resolution sequence

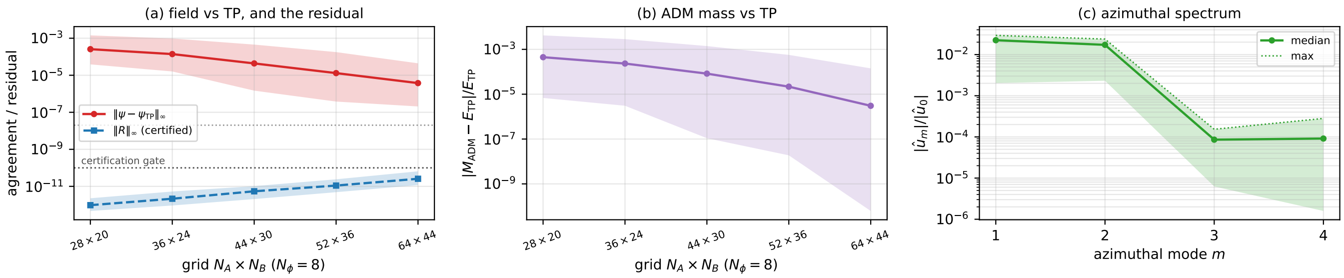

FIG. 8. **Quasi-circular data validated against TwoPunctures, across the production box.** Each curve is a median over 100 quasi-circular configurations drawn from the production box by a Latin hypercube, with the shaded envelope spanning the sample minimum to maximum; the resolution axis is the meridional sequence at fixed $N_\phi = 8$, and every configuration is compared with one TwoPunctures solve at $72 \times 72 \times 12$. *(a)* The pointwise field difference $\|\psi - \psi_{\mathrm{TP}}\|_\infty$ falls with resolution while the certified Newton–Krylov residual $\|R\|_\infty$ rises. The opposite trends are the point: the rise is roundoff amplification in unpopulated high-m modes (Sec. VII), not a loss of convergence, and every configuration stays below the certification gate 10^{-10} (dotted) at every resolution. Here $\|R\|_\infty$ is the row-equilibrated residual of Sec. III B, evaluated at the returned iterate. *(b)* The integral comparison, $|M_{\mathrm{ADM}} - E_{\mathrm{TP}}|/E_{\mathrm{TP}}$. *(c)* The azimuthal Fourier spectrum $|\hat{u}_m|/|\hat{u}_0|$ at the finest resolution, which sets how many ϕ modes the quasi-circular data require.

for each, at fixed $N_\phi = 8$; every configuration is compared with a single TwoPunctures solve at $72 \times 72 \times 12$, which out-resolves the finest of our own grids in every direction. Latin hypercube sampling rather than independent uniform draws keeps the one-dimensional marginals stratified, which matters in eight dimensions; we additionally solve ten configurations placed deliberately at the box edges, reported separately so that they do not bias the sample statistics. Refining at fixed parameters is what identifies the residual disagreement as a discretization error, so the ladder is retained and the sample enters as the band around it.

The pointwise field difference converges across the sampled box, as shown in Fig. 8(a). At the coarsest resolution the sample spans 3.9×10^{-5} to 1.4×10^{-3} with a median of 2.6×10^{-4}; at the finest it spans 2.1×10^{-7} to 4.4×10^{-5} with a median of 3.8×10^{-6}, a median improvement of a factor of 68. The minimum, median and maximum of the sample each fall monotonically along the sequence, which is the statement the band makes. Individual configurations need not: 13 of the 100 show one non-monotone step somewhere along the ladder, as a supremum norm of a spectral error may, and we therefore claim convergence of the distribution rather than of every member of it. The disagreement is limited by spatial resolution rather than by the nonlinear solve, and it is the meridional resolution that limits it: raising N_ϕ alongside (N_A, N_B) leaves the field difference unchanged to three digits at every resolution. Two parameters control where a configuration sits within the band. At fixed spin the difference grows by roughly two orders of magnitude from $q = 1$ to $q = 3$, and by a further order from $b = 3$ to $b = 10$; nonspinning configurations reach the 10^{-10} range. Spin combined with an unequal mass ratio, not either alone, is therefore what sets the scale of the comparison.

The certified residual behaves in the opposite direction, its median rising by a factor of 27 along the same sequence, from 9.7×10^{-13} to 2.6×10^{-11}, while remaining below the certification tolerance 10^{-10} for every one of the 100 configurations at every resolution; the largest value anywhere in the sample is 6.5×10^{-11}. The two trends together are what identifies the rise as roundoff amplification in unpopulated high-m modes rather than a loss of convergence, since the field difference falls while the residual grows. The direction of refinement decides the size of the effect: raising N_ϕ instead makes the residual rise by six orders of magnitude and breach the tolerance, which is why the production sequence refines the meridian.

The integral comparison, shown in Fig. 8(b), converges faster than the pointwise one: the median relative difference in the ADM mass falls from 4.5×10^{-4} to 3.0×10^{-6}, a factor of 147, and the best configuration reaches 6.2×10^{-11}. It would be natural to expect the integral to be uniformly the more accurate of the two, being determined by the asymptotic behavior of the conformal factor and so insensitive to the local high-frequency error that limits a pointwise comparison. Across this box it is not. At the finest resolution the two are comparable in the median, their ratio being 1.1, and the ADM mass is the more accurate for only about half of the sample. What decides the ordering is the separation rather than the spin: at $b = 3$ the integral is between two and twenty-five times the more accurate, while at $b = 10$ it is two to three times the less accurate, its boundary derivative at $A = 1$ being read from an increasingly small $\psi - 1$. Nonspinning configurations fall on both sides. Both metrics are therefore worth reporting, and neither substitutes for the other.

The azimuthal Fourier spectrum in Fig. 8(c) sets the truncation N_ϕ requires. The quasi-circular data are non-axisymmetric by construction: the tangential momentum populates $m = 2$ and generic spins populate $m = 1$, and across the sample both sit at the percent level, with medians of 1.7×10^{-2} and 2.2×10^{-2} and maxima of 2.4×10^{-2} and 2.9×10^{-2} respectively. The spectrum then falls by more than two decades, the $m = 3$ and $m = 4$ medians being 8.6×10^{-5} and 9.1×10^{-5}. A small number of Fourier modes therefore resolves these data, and the quasi-circular family exercises the Fourier-in-ϕ solver of Sec. III A on its own.

Two checks that the quasi-circular family cannot supply, because it is never axisymmetric, complete the picture. First, for a head-on configuration with spin along the collision axis, all $m \geq 1$ modes remain at most 9.2×10^{-17} across mass ratios and spin magnitudes; this is an internal consistency check on the mode decoupling of the Fourier-in-ϕ operator, verifying that it does not manufacture spurious non-axisymmetry. Second, in this axisymmetric limit the pointwise field agreement approaches machine precision: for an equal-mass configuration with $b = 3$ and axial momentum $P = 0.5$, solved on the same grid sequence and compared against the same reference, the two independently discretized spectral codes agree in ψ to 2.6×10^{-10} in the supremum norm, with a root-mean-square difference of 4.3×10^{-11}, and in the total ADM mass to a relative difference of 5.0×10^{-11}; the certified residual there is 1.2×10^{-12}. With no azimuthal structure to limit either code, this is the most stringent code-to-code reference available for the solver, some four orders of magnitude below the median of the quasi-circular band.

All pointwise differences quoted here are estimated on a fixed set of 69 interior probe points, placed clear of both punctures and covering the full azimuthal range. The density matters, and not uniformly. A sparse probe set underestimates a supremum norm preferentially where the field is best resolved, because the estimate collapses whenever the error at the dominant point changes sign; on coarse grids a sparse and a dense set agree, so the bias falls precisely on the converged values one would quote. Reducing the set to six points lowers the difference at the finest resolution by a median factor of 2.7 across the sample and by as much as 67 for individual configurations, and on the axisymmetric configuration above by a factor of 16. The reference solve is converged well below every difference reported: its field moves by 9×10^{-16} between its own 72×72 and 96×96 meridional resolutions, and its azimuthal truncation contributes at most 2×10^{-8}, which is two orders of magnitude below the field difference at the finest resolution for the configurations on which it was measured. The comparison therefore reflects our discretization rather than the reference.

The ADM angular momentum provides a further integral diagnostic, computed from the conformal Bowen–York tensor by a York surface integral and compared both with the closed-form expression

$$J^i = \sum_X \left[(S^X)^i + (x_X \times P_X)^i \right]$$

and with TwoPunctures. For pure spin, the inclination of the ADM angular momentum relative to the collision axis agrees with the spin inclination to machine precision for all tested spin magnitudes and separations. An antisymmetric off-axis momentum P_x produces the expected orbital contribution $J_y = 2bP_x$, which the surface integral recovers. After rotation to a common frame, the angular momentum reported by TwoPunctures agrees to $\leq 1.4 \times 10^{-14}$.

These results validate both the three-dimensional solver and the quasi-circular data used by the parametric models, over the box the models are built on rather than at a single configuration. The comparison with TwoPunctures provides an external cross-check of the numerical implementation; constraint satisfaction of the final data is established independently by the certified residual described in Sec. IV C.

2. Constraint violation on an evolution grid

As a final validation, we interpolate the spectral initial data onto a uniform Cartesian grid, as required by a numerical-relativity evolution code, and evaluate the Hamiltonian and momentum constraints using a generic finite-difference constraint monitor. The monitor computes the Christoffel symbols and Ricci tensor directly from finite differences, without making use of conformal flatness, and therefore provides a test independent of the spectral discretization used to construct the initial data.

We use the axisymmetric configuration of the preceding section for this test. What it measures is a property of the Cartesian discretization rather than of any particular slice, so the simplest configuration is the appropriate one: it carries the fewest confounds, and the axisymmetric limit is where the two initial-data solutions differ least, which is what makes the comparison below sharp. For that equal-mass configuration both constraint norms converge at the expected second order with Cartesian grid spacing h. The measured convergence orders are 2.14 for $\|\mathcal{H}\|_2$ and 2.02 for $\|\mathcal{M}\|_2$. Repeating the calculation with the conformal factor obtained from TwoPunctures gives identical constraint violations on the same Cartesian grids. The measured residuals are therefore set by the finite-difference truncation error of the constraint monitor rather than by differences between the two initial-data solutions.

This test shows that, after interpolation onto an evolution grid, the constraint violations of our initial data are indistinguishable from those obtained from TwoPunctures at the same finite-difference resolution. The accuracy of the spectral initial data therefore lies below the truncation error introduced by the Cartesian discretization used in this test.

TABLE II. **The tangent of Eq. (23) against the certified solve.** Maximum relative difference between the implicit-function-theorem tangent $\partial U/\partial\theta_k$, obtained from one back-solve against the factored Jacobian, and second-order central finite differences of the certified Newton–Krylov solve with step h. Non-axisymmetric quasi-circular slice at $b = 2.4$, $q = 1.7$, $P_t = 0.5$, $\chi_y^A = 0.30$, $\chi_y^B = 0.20$, on the $(N_A, N_B, N_\phi) = (18, 14, 6)$ grid; both solves are certified to $\|R\|_\infty = 2.4 \times 10^{-14}$. The two tangents agree to the roundoff floor of the solve, so the comparison is quoted as an order-of-magnitude bound: its mantissa depends on the order of floating-point reductions, and therefore on the hardware and the linear-algebra library, whereas the exponent does not.

Axis θ_k	h	Eq. (23) vs. finite differences
b	10^{-4}	$< 10^{-8}$
q	10^{-4}	$< 10^{-8}$
χ_y^A	10^{-4}	$< 10^{-8}$
χ_y^B	10^{-4}	$< 10^{-8}$

Appendix B: Verification of the parameter sensitivities

We verify the parameter sensitivities at two levels. First, we test whether Eq. (23) gives the sensitivity of the certified elliptic solve. We then test whether the gradient exposed by the parametric model reproduces this sensitivity.

Table II tests the first statement on the production three-dimensional operator by comparing the implicit tangent of Eq. (23) with central finite differences of independently certified Newton–Krylov solutions. The two agree to $\sim 10^{-9}$ along every axis of the aligned-spin parameter domain, at the accuracy of the finite-difference reference. This verifies that the analytic $\partial R/\partial\theta_k$ and the subsequent back-solve reproduce the parameter derivative of the certified solution.

We next verify the sensitivity exposed by the parametric model. Table III compares its automatic-differentiation gradient with two independent references. First, automatic differentiation agrees with central finite differences of the same model to 10^{-10}–10^{-9}, verifying the differentiation of the interpolant. Second, and more importantly, the exposed gradient agrees with the independently computed implicit tangent of Eq. (23) to $\sim 2 \times 10^{-7}$ along the spin axes and $\sim 10^{-5}$ along b and q. The remaining difference is set by the interpolation error of the derivative. It is largest along b and q, consistent with the slower field interpolation along these directions discussed in Sec. V B, and decreases exponentially with increasing parameter order.

The two tests therefore probe distinct parts of the differentiable pipeline. Table II verifies the tangent of the underlying certified solve, while Table III verifies that differentiation of the model recovers this tangent to the accuracy of the parametric interpolation.

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TABLE III. **The exposed sensitivity against both references.** Maximum relative difference between the automatic-differentiation sensitivity $\partial U/\partial\theta_k$ of the parametric model and, respectively, second-order central finite differences of the same model with step h, and the implicit-function-theorem tangent of Eq. (23) computed independently from the certified solve at the same parameter point. Two-centre aligned-spin interpolants on the $(N_A, N_B) = (36, 24)$ grid at $M = 1$: a (b, q) model of order $Q = (12, 10)$ over $b \in [3, 12]$, $q \in [1, 3]$, evaluated off-node at $(b, q) = (6.337, 1.713)$; and a (χ_y^A, χ_y^B) model of order $Q = 8$ over $[0, 0.6]^2$ at fixed $q = 1.5$, $b = 4$, evaluated off-node at $(\chi_y^A, \chi_y^B) = (0.413, 0.227)$, where χ_y^A and χ_y^B are the aligned spin components on the two punctures. The reference solves are certified to $\|R\|_\infty \leq 1\times10^{-11}$. The finite-difference column is quoted as an order-of-magnitude bound because it sits at the roundoff floor of the solve: the two estimates agree to some nine digits, so their difference measures the accumulated rounding rather than any modelling error, and its mantissa moves with the order of floating-point reductions. The comparison against Eq. (23) in the last column is between two analytic quantities, lies far above that floor, and reproduces exactly.

Axis θ_k	h	vs. finite differences	vs. Eq. (23)
b	10^{-6}	$< 10^{-8}$	1.3×10^{-5}
q	10^{-6}	$< 10^{-9}$	1.2×10^{-5}
χ_y^A	10^{-6}	$< 10^{-9}$	1.7×10^{-7}
χ_y^B	10^{-6}	$< 10^{-9}$	1.8×10^{-7}

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